Republic of the Philippines
ENERGY REGULATORY COMMISSION
Pasig City

ENERGY REGULATORY COMMISSION
APPROVED FOR POSTING
www.erc.gov.ph

RESOLUTION NO. 11, SERIES OF 2021

A RESOLUTION ADOPTING THE REGULATORY ASSET BASE (RAB) ROLL-FORWARD HANDBOOK FOR PRIVATELY-OWNED ELECTRICITY DISTRIBUTION UTILITIES (DUS)

WHEREAS, Section 43 (f) of Republic Act No. 9136, otherwise known as the Electric Power Industry Reform Act of 2001 (EPIRA) and 5 (a), Rule 15 of its Implementing Rules and Regulations (IRR), authorize the Commission to adopt alternative forms of internationally accepted rate-setting methodology;

WHEREAS, on 10 December 2004, the Commission issued Resolution No. 12-02, Series of 2004 entitled, *"Adopting a Methodology for Setting Distribution Wheeling Rates,"* wherein it employed a performance-based rate setting methodology for setting distribution wheeling rates of privately owned distribution utilities embodied in a document denominated as the "*Guidelines on the Methodology for Setting Distribution Wheeling Rates,"* or the *"Distribution Wheeling Rates Guidelines" (DWRG);*

WHEREAS, on 11 April 2006, the Commission issued Resolution No. 16, Series of 2006 entitled, *"Adopting a Policy that Entry to the Performance Based Regulation (PBR) by Private Utilities under the Distribution Wheeling Rates Guidelines (DWRG) be Made Mandatory"*;

WHEREAS, on 26 July 2006, the Commission issued Resolution No. 39, Series of 2006 entitled, *"Adopting the Rules for Setting the Distribution Wheeling Rates (RDWR) for Privately Owned Distribution Utilities Entering Performance Based Regulation (First Entry Point)"*;

WHEREAS, on 13 December 2006, the Commission issued Resolution No. 54, Series of 2006 entitled, *"Adopting the Rules for Setting the Distribution Wheeling Rates (RDWR) for Privately Owned Distribution Utilities Entering Performance Based Regulation (Second and Later Entry Points)"*;

WHEREAS, on 08 December 2008, the Commission issued Resolution No. 20, Series of 2008 entitled, *"Resolution Modifying the Rules for Setting Distribution Wheeling Rates (RDWR) for Privately-Owned Distribution Utilities Entering Performance Based Regulation (RDWR),"* wherein it adopted the RDWR for Third Entry Group entering PBR for the Second Regulatory Period";

WHEREAS, on 12 July 2016, the Commission issued Resolution No. 25, Series of 2016 entitled, *"Resolution Modifying the Rules for Setting Distribution Wheeling Rates (RDWR) for all Privately Owned Distribution Utilities Entering Performance Based Regulation,"*[1] wherein it adopted the modified RDWR to be applicable to all privately-owned distribution utilities operating under the Performance-Based Regulation (PBR);

WHEREAS, on 21 November 2016, the Commission issued a *Notice* dated 15 November 2016 under ERC Case No. 2016-003 RM,[2] wherein it approved for posting in its official website (www.erc.gov.ph) the *Draft Regulatory Asset Base (RAB) Roll Forward Handbook for Privately Owned Electricity Distribution Utilities* (First Draft RAB Roll Forward Handbook) and directed the submission of comments by any interested stakeholders until 19 December 2016. In the same *Notice,* the Commission set the public consultations on 06 January 2017 and 09 January 2017.

WHEREAS, on various dates, the Commission received written comments from the following stakeholders: (1) Cagayan Electric and Light Company (CEPALCO); (2) Dagupan Electric Corporation (DECORP); (3) Manila Electric Company (MERALCO); and (4) Mr. Uriel Borja;

---

[1] Resolution No. 25, Series of 2016 did not become effective due to non-publication within the required period and the reset process pursuant to the said Resolution had already lapsed and need to be revised.

[2] Entitled, "In re: Draft Regulatory Asset Base (RAB) Roll Forward Handbook for Privately Owned Electricity Distribution Utilities (DUs)".

WHEREAS, on 03 May 2021, the Commission issued another *Notice* dated 29 April 2021 wherein it approved for posting the 2nd draft *RAB Rolled-Forward Handbook* and directed the submission of comments thereto by any interested stakeholders until 18 May 2021.

WHEREAS, on various dates, the Commission received various written comments to the 2nd draft *RAB Rolled-Forward Handbook* from the following stakeholders: (1) Private Electric Power Operators Association (PEPOA); (2) MERALCO; (3) Cotabato Light and Power Company (CLPC); (4) Visayan Electric Company (VECO); (5) Davao Light and Power Company (DLPC); and (6) Subic Enerzone Corporation (SEZC);

WHEREAS, on 02 December 2021, the Commission approved the Revised Rules for Setting Distribution Wheeling Rates (RDWR) docketed under ERC Case No. 2019-003 RM;[3]

WHEREAS, Section 4.8.1 of the Revised RDWR describes the requirements for the determination of the RAB value for each DU, which shall either be through (a) the revaluation of a DU's RAB, or (b) the roll forward of the DU's previous RAB value;

WHEREAS, after careful evaluation of all the comments and submissions of the stakeholders, and those made during public consultations, the Commission, on 02 December 2021, approved the RAB Roll-Forward Handbook for privately-owned DUs operating under PBR for their subsequent Regulatory Periods;

NOW, THEREFORE, **BE IT RESOLVED**, as the Commission hereby **RESOLVES** to **APPROVE** and **ADOPT** the *Regulatory Asset Base (RAB) Roll-Forward Handbook for Privately-Owned Electricity Distribution Utilities (DUs),* hereto attached and made an integral part of this *Resolution* as Annex "A".

[3] Entitled, "In Re: Rules for Setting Distribution Wheeling Rates (RDWR)".

This *Resolution* shall take effect immediately following its publication in a newspaper of general circulation, or in the Official Gazette.

Let copies of this *Resolution* be furnished all parties concerned, the University of the Philippines Law Center – Office of the National Administrative Register (UPLC-ONAR), all privately-owned Distribution Utilities, as well as published in the ERC website and such other online platform available to the Commission.

Pasig City, 02 December 2021.

AGNES VST DEVANADERA
Chairperson and CEO

ERC
Office of the Chairperson
AVSTD-2021-12-560-0002

ALEXIS M. LUMBATAN
Commissioner

*Voted Nay**
CATHERINE P. MACEDA
Commissioner

FLORESINDA G. BALDO-DIGAL
Commissioner

MARKO ROMEO L. FUENTES
Commissioner

LS: BJVG/LSP/MCCG

ROS: ABS/AJMO/LLG

*Commissioner Maceda Voted Nay for the following reasons:
No discussion on changes introduced in the final RDWR, the RAB Handbook, and other documents during the Commission Meeting. Separate Explanation on Dissenting / Negative Vote submitted to form part of the records of the Commission.

Regulatory Asset Base (RAB) Roll Forward Handbook for Privately Owned Electricity Distribution Utilities (DUs)

Final
December 2021

TABLE OF CONTENTS

Table of Contents *2*

1. Introduction *4*

1.1 Overview 4

1.2 ERC Position on the Roll Forward Process 4

1.3 Amendment 5

1.4 Effectivity 5

1.5 Definitions 5

2. Roll Forward Process Requirements *10*

2.1 Timeframe of Submission 10

2.2 Required INDEPENDENT Expert Opinions 10

2.3 Independent Expert's Report Requirements 10

2.4 Required Information for the Directors' Certificate 12

2.5 Review of DU Submitted information and Reports 13

3. Roll Forward Variables *15*

3.1 Introduction 15

3.1.1 Allowed Assets in the RAB 15

3.1.2 RAB Asset Register 16

3.1.3 Asset Categories 17

3.1.4 Description of Asset Categories and Asset Classes 20

3.2 Opening RAB for Regulatory Year 1 23

3.3 Asset Additions for Current Regulatory Period 23

3.4 Asset Disposals for Current Regulatory Period 23

3.5 Depreciation 23

3.5.1 Standard Asset Lives 24

3.5.2 Asset Lives for Refurbished Assets 27

3.5.3 Remaining Lives 27

3.5.4 Residual Values 27

3.6 Excluded Variables 28

4. Roll Forward Procedures *29*

4.1 Introduction 29

4.2 Roll Forward Procedure 29

4.2.1 Adjusting Asset Additions, Asset Disposals and Depreciation for the Previous Regulatory Period 30

Actual disposals are defined as the Asset Disposal value minus the value obtained from the asset's disposal. 31

4.3 Optimization 31

4.3.1 Introduction 31

4.3.2 Optimization Constraints 31

4.3.3 Quality of Supply 32

4.3.4 Future Load Growth 33

4.3.5 The Process of Optimization 35

5. List of Acronyms and Shortcuts *37*

Appendix A: Roll Forward Template Samples *38*
Appendix B: Optimization *40*
Appendix C: Updated Replacement Costs *48*
Standard Replacement Costs for Overhead LINES: Conductors **52**
Standard Replacement Costs for Overhead Lines: Poles **53**
Standard Replacement Costs for Overhead Lines: Pole Top Hardware **54**
Standard Replacement Costs for Customer Service Connections **55**
Standard Replacement Costs for Meters and Metering Accessories **56**
Standard Replacement Costs for Overhead Line Devices **57**
Standard Replacement Costs for Switches and Circuit Breakers **58**
Standard Replacement Costs for Power Transformers **59**

1. INTRODUCTION

1.1 OVERVIEW

The Energy Regulatory Commission (ERC/Commission) has promulgated the "Rules for Setting Distribution Wheeling Rates (RDWR) for Privately Owned Electricity Distribution Utilities (DUs) Operating under Performance Based Regulation (PBR)"[1]. Section 4.8.1 of the RDWR describes the requirements for the determination of the Regulatory Asset Base (RAB) value for each DU, which shall either be through (a) the revaluation of a DU's RAB or (b) the roll forward of the DU's previous RAB value.

This Roll Forward Handbook (Handbook) details the ERC's roll forward process which would apply to all DUs operating under PBR for the subsequent Regulatory Period. Consistent with the requirements in the RDWR, DUs are required to submit a proposed RAB value as part of their rate applications. This Handbook will be used to determine the Closing RAB for each DU. This Closing RAB value then becomes the Opening RAB to be used as an input parameter for the building block model used by the ERC to calculate the annual revenue requirement (ARR) for each DU for the next Regulatory Period.

1.2 ERC POSITION ON THE ROLL FORWARD PROCESS

At the time of writing of this Handbook, the position of the ERC is that there are strong economic arguments and support in the submissions for the roll forward approach.

The basic steps involved in the roll forward process as envisioned by the ERC[2] are summarized in the table below as extracted from Section 5.1.1 of the Position Paper[3] of the Commission. The detailed methodology in the application of the roll forward approach is described in more detail in the other sections of this Handbook[4].

Table 1: ERC's Steps in Determining the Roll Forward Asset Values[5]

Opening RAB	Adopt the latest approved Opening RAB
Add: Actual Capital Expenditure (CAPEX)	Optimized actual cost of new assets or investment that has been added
Less: Regulatory Depreciation	Actual return of capital allowed under the RDWR (based on the standard asset lives set by the ERC)

[1] ERC Case No. 2019-003 RM Annex "A", "Rules for Setting Distribution Wheeling Rates for Privately Owned Electricity Distribution Utilities Operating Under Performance Based Regulation", Energy Regulatory Commission, December 2021.

[2] The RAB Handbook was simultaneously amended in consideration of the latest RDWR and the Position Paper issued in December 2021.

[3] "Position Paper: Regulatory Reset for the July 2022 to June 2026 Fifth Regulatory Period for the First Entry Group of Privately Owned Distribution Utilities subject to Performance Based Regulation", Energy Regulatory Commission, December 2021.

[4] The formula for the calculation of the Opening RAB is discussed in more detail in Section 4, Asset Additions are described in more detail in Section 3.3, Asset Disposals are discussed in more detail in Section 3.4 and depreciation is discussed in more detail in Section 3.5.

[5] The Commission excludes the addition of inflation factors at the time of roll forward RAB calculation which is subsequently excluded from Section 4. The Commission shall consider the application of any inflation factor only once, either thru the RAB or WACC.

Less: Asset Disposal/Retirement (net income from disposal)	Optmized actual cost of disposed/retired assets less net income from disposal
Closing RAB	Closing value of the RAB to be carried forward at the start of the next Regulatory Period

1.3 AMENDMENT

This Handbook may be updated by the ERC. Nevertheless, any update thereto shall be made only after every regulatory period and not within the regulatory period. An updated version will be published by the ERC when required.

1.4 EFFECTIVITY

This Handbook supersedes the "Valuation Handbook for Optimized Depreciated Replacement Cost Valuation of System Fixed Assets of Privately Owned Distribution Utilities Operating Under Performance-Based Regulation (Third Regulatory Period)",[6]as well as the "Asset Valuation Policy Guidelines for Privately Owned Distribution Utilities Subject to Performance Based Regulation", Energy Regulatory Commission, August 2006",[7] and will take effect following its publication or posting.

1.5 DEFINITIONS

TERM	DEFINITION
Actual Nominal	Actual values (excluding adjustments for inflation) incurred for each regulatory year over the Regulatory Period.
Ancillary Services	Those services that are necessary to support the transmission of capacity and energy from resources to loads while maintaining the reliable operation of the Grid or a Distribution System or a Subtransmission System in accordance with good utility practice, the Grid Code and the Distribution Code. These include the services rendered by generation facilities that are dedicated to providing emergency standby power during emergency situations, such as when the normal operating conditions of a Distribution System are compromised due to bulk supply factors beyond the control of a Regulated Entity.
Asset Additions	Refers to the actual or forecast Capital Expenditure of a Distribution Utility under the different Asset Categories for the period from the date of the valuation of the Distribution Utility in the previous Regulatory Period to the date of the commencement of the subsequent Regulatory Period.
Asset Category	A grouping of assets in the asset base that perform a similar function. The different Asset Categories are identified in Section 3.1.3.
Asset Class	Refers to the asset breakdown within an asset category. The different Asset Classes are identified in Section 3.1.4.

[6] "Valuation Handbook for Optimized Depreciated Replacement Cost Valuation of System Fixed Assets of Privately Owned Distribution Utilities Operating Under Performance-Based Regulation (Third Regulatory Period)", Energy Regulatory Commission, July 2010.
[7] "Asset Valuation Policy Guidelines for Privately Owned Distribution Utilities Subject to Performance Based Regulation", Energy Regulatory Commission, August 2006

TERM	DEFINITION
Asset Disposals	Refers to the actual or forecast net receipts from the disposal of assets, as the case may be, for the period from the date of the valuation of the Distribution Utility in the previous Regulatory Period to the date of the commencement of the subsequent Regulatory Period.
Asset Type	Refers to a sub-category of assets within an asset class.
Brownfields	The 'Brownfields' cost basis assumes construction occurs for a new electricity network in a particular area, but installed around existing infrastructure and developments, with the bulk electricity supply to the area drawn from the most efficient existing bulk supply points in the vicinity of the area. (Such points can be an existing Point of Supply, or access point to a DU's existing electricity bulk-supply network.) By comparison, the 'Greenfields' cost basis assumes that construction occurs in an area free of any prior development, where the bulk electricity supply would be drawn from an optimal access point, the location of which would be part of the overall installation.
Closing RAB	Closing value of the Regulatory Asset Base of the Distribution Utility to be carried forward at the start of the next regulatory year or the next Regulatory Period.
Consumer Price Index	Refers to the "The All Items Consumer Price Index" published by the Philippine Statistics Authority (PSA).
Distribution Connection Services	In respect of a regulated Distribution System: a. the provision of capability at a distribution connection point in respect of that regulated Distribution System to deliver electricity to or take electricity from that distribution connection point; b. the conveyance of electricity: i. to a distribution connection point in respect of that regulated Distribution System from any user system or equipment which is directly connected to that regulated Distribution System at that distribution connection point; or ii. from a distribution connection point in respect of that regulated Distribution System to any user system or equipment which is directly connected to that regulated Distribution System at that distribution connection point; or iii. from an embedded generator to a regulated Distribution System, also where the generation plant is installed outside the franchise area of the Distribution Utility operating the regulated Distribution System. c. the planning, installation, maintenance, augmentation, testing and operation of distribution connection assets in respect of that regulated Distribution System; and d. the provision of services that support any of the services referred to in paragraphs a to c.
Distribution Substation	A substation that transforms subtransmission voltages to primary distribution voltages, for example, 69/23kV and 115/34.5kV substations.
Distribution System	In respect of a Distribution Utility, a system of wires and associated facilities extending between the delivery points on the grid and any sub-transmission system operated by a person other than the Distribution Utility, on the one hand, and the points of connection of user systems and equipment of end-users, on the other hand.

TERM	DEFINITION
Distribution Utility	In respect of this Handbook, a private corporation that has an exclusive franchise to operate a Distribution System in accordance with the EPIRA.
Energy Regulatory Commission	The Energy Regulatory Commission created by Section 38 of the EPIRA.
EPIRA	Republic Act No. 9136, otherwise known as the Electric Power Industry Reform Act of 2001.
Equipment	All apparatus, machines, conductors, etc. used as a part of, or in connection with, an electrical installation, as defined in the Distribution Code.
Expert	Means a person(s) which possess all the following qualifications: a. Is a recognized engineer(s) qualified to be appointed as an expert in regulatory valuation, or is an engineering/consultancy/appraisal company recognized as being qualified to act as an expert and represented by a recognized engineer(s) qualified to be an expert in regulatory valuation, and has manpower and resources to enable the performance of the necessary work;Who has adequate manpower and resources to enable the performance of the necessary work; b. Is acting in the professional capacity as specified above; c. Has knowledge with the Republic Act 9136 or the Electric Power Industry Reform Act (EPIRA) and its Implementing Rules and Regulations (IRR), or of the Utility rate setting; d. Has knowledge with the ERC's issuances such as the Rules for Setting Distribution Wheeling Rates (RDWR), the RAB Handbook, the Issues Paper and the Position Paper; e. Has previous exposure to regulatory valuation in utility rate setting or regulatory reset process for the last ten (10) years; and at least five (5) years of experience in undertaking projects in utility rate setting or regulatory reset process; f. Has been involve in the regulatory valuation of utilities in the Philippines or in other jurisdictions; knowledgeable with the different methodology used in regulatory valuation or with the roll-forward RAB approach to properly provide an Independent Expert's Report as required in this document; g. Has the ability to abide by the timeframe and schedule; and h. Is not related within the fourth civil degree of consanguinity or affinity to any of the directors, trustees, officers or their spouses of the subject DU; that the Expert is not a stockholder of the subject DU; and that there is no existing relationship with, or interest in the Distribution Utility to be reviewed that may involve the person in a conflict of interest between his, her or its duties to the Distribution Utility and his, her its duties to the Energy Regulatory Commission.

TERM	DEFINITION
Excluded Service	A service provided in a qualified franchise area in the ordinary course of an electricity distribution business that is neither a Regulated Distribution Service nor a service that is contestable (whether or not a service is contestable will be determined by the ERC).
First Closing Date	Means the closing date for the latest approved Regulatory Asset Base of a Distribution Utility, i.e. for the first entry point of Distribution Utilities the First Closing Date is June 30, 2015, for the second entry point of Distribution Utilities the First Closing Date is March 31, 2013 and so on.
Incremental optimization	An approach in which the existing network is examined and changes made to ratings, configurations, designs, or materials to optimize the network configuration and its component asset having regard to such issues as excess capacity, redundancy, and over-design.
Independent Expert's Report	Means a report prepared and issued by a person or firm that is qualified to be an Expert, details of the report are specified in Section 2.3 of this document.
Modern Equivalent Asset	Electrical equipment that would be installed at the current date, replacing older assets that are already in place and would fulfill, at the minimum, the same function of those being replaced. Modern Equivalent Asset would be those widely accepted by efficient Distribution Utilities to be the most technically and economically effective assets currently available to fulfill a particular function that can be practically installed in the field by Distribution Utilities.
Network Assets	Refers to assets forming part of the Regulatory Asset Base that are required to provide Regulated Distribution Services, Distribution Connection Services, and Regulated Retail Services but are not Non-Network Assets.
Non-Network Assets	Those assets forming part of a Regulatory Asset Base that are required to provide Regulated Distribution Services, but are not Distribution System assets or Distribution Connection assets. Examples include construction vehicles and equipment, office buildings, depots, warehouses, furniture and computers used for non-network related applications.
Opening RAB	Opening value of the Regulatory Asset Base of the Distribution Utility for the regulatory year or for the Regulatory Period.
Regulated Distribution Services	In respect of a regulated Distribution System: a. the conveyance of electricity through the regulated Distribution System and the control and monitoring of electricity as it is conveyed through the regulated Distribution System (including any services that support such conveyance, control or monitoring or the safe operation of the regulated Distribution System); b. the planning, maintenance, augmentation and operation of the regulated Distribution System; c. the provision, installation, commissioning, testing, repair, maintenance and reading both of meters that are used to measure the delivery of electricity to persons whose user systems or equipment is directly connected to the regulated Distribution System and of other meters that are used (for the purposes of the Wholesale Electricity Spot Market) to measure the flow of electricity into or through the regulated Distribution System;

TERM	DEFINITION
	d. Distribution Connection Services in respect of the regulated Distribution System except to the extent that such Distribution Connection Services have been determined by the Energy Regulatory Commission to be excluded services under Section 1.6 of the RDWR; e. the provision of ancillary services that are provided using assets which form part of the regulated Distribution System (excluding any such ancillary services to the extent they are provided to the system operator under contract, or through a spot market established under the Wholesale Electricity Spot Market Rules); and f. billing, collection and customer services that are directly related to the delivery of electricity through the regulated Distribution System to distribution connection points in respect of the regulated Distribution System and billing, collection and customer services for persons purchasing or receiving (or seeking to purchase or receive) any Distribution Connection Services in respect of the regulated Distribution System, but do not include such of these services as are determined by the Energy Regulatory Commission to be contestable.
Regulated Retail Services	The distribution business segment defined in the Business Separation Guidelines[8] that relates to the provision of retail services pertaining to the sale of electricity to end users who are included in the captive market.
Regulatory Asset Base	Those assets employed by a Distribution Utility to provide efficient regulated distribution services. It covers the regulated distribution network assets as well as the Non-Network Assets required to support the delivery of Regulated Distribution Services.
Regulatory Period	The First Regulatory Period, the Second Regulatory Period or a subsequent Regulatory Period (as the case may be).
Roll Forward Information	Refers to all information provided by the Distribution Utility for purposes of the roll forward.
Roll Forward Template	Refers to the excel spreadsheet template that will be used by each Distribution Utility in its roll forward calculation. The template has two (2) main worksheets, the Opening RAB template and the Closing RAB template.
System Fixed Assets	Refer to assets forming part of the Regulatory Asset Base that are required to provide Regulated Distribution Services.

[8] The Guideline promulgated by the Energy Regulatory Commission under Rule 10, Section 1 of the Implementing Rules and Regulations as promulgated by the Energy Regulatory Commission on September 2003 under ERC Case No. 2003-46, Resolution No. 49, Series of 2006 as amended, to provide the framework and rules for the structural unbundling of the business activities of electric power industry participants.

2. ROLL FORWARD PROCESS REQUIREMENTS

2.1 TIMEFRAME OF SUBMISSION

Unless otherwise specified by the ERC, the determination of the roll forward RAB value must be undertaken for each DU, which will commence with the submission of the following:

a. Independent Expert's Report complying with requirements set out in this Handbook; and

b. Directors' Certificate to be completed and signed by a minimum of two board of directors of the DU.

The Independent Expert's Report are to be completed before the start of the next Regulatory Period or as otherwise specified by the ERC. The Directors' Certificate is to be submitted together with the Independent Expert's Report. Failure to comply with the requirements provided herein shall be a ground for the ERC to reject the submission of the DU.

2.2 REQUIRED INDEPENDENT EXPERT OPINIONS

It will be each DU's responsibility to appoint the required experts as defined in this document.

2.3 INDEPENDENT EXPERT'S REPORT REQUIREMENTS

The Independent Expert's Report must be completed by an Expert as defined in this document. The report at a minimum must:

a. Include a copy of the appointment of the Expert by the DU detailing the scope of work of the Expert;

b. Present the results of the optimization process[9] undertaken in accordance with the requirements of this Handbook, including the review of the DU's records concerning asset utilization;

c. Include a copy of the quality of supply criteria for all period under review used by the DU for planning purposes;

d. Include a section detailing the adjusted Asset Additions, Asset Disposals and depreciation for the previous Regulatory Period, if relevant.[10] The section shall be accompanied by the following:

   i. Detail regarding the previous Regulatory Period's approved forecast versus actual Asset Additions, Asset Disposals and depreciation including sufficient detail on Asset Additions (and any subsequent Asset Disposals and depreciation) not originally part of the forecast for the previous Regulatory Period but approved by the ERC as emergency CAPEX;

   ii. Review of the DU's records in relation to the timing of assets added or removed over the period under review;

[9] Refer to Section 4.3
[10] Refer to Section 4.2.1.

iii. Assessment and verification of installation dates[11] of Asset Additions over the period under review;

iv. Verification of the calculation of the Asset Disposals to include but not limited to income derived from the disposal and depreciated value, and validation of the asset disposed;

v. Detail regarding the variables (dates, costs, standard asset lives, remaining lives, etc.) used in the prescribed formula, including the source of information and level of confidence in the accuracy of the variables used;

vi. An excel spreadsheet clearly identifying the calculation from which the updated previous Regulatory Period Closing RAB is calculated; and

vii. Completed Roll Forward Template as prescribed by the Commission, specifically the template for the Opening RAB.

e. Include a section detailing the development of the Closing RAB of the current Regulatory Period.[12] The section shall be accompanied by the following:

i. Review of the DU's records in relation to the timing of assets being added or removed;

ii. Assessment and verification of installation dates of Asset Additions;

iii. Verification of the calculation of Asset Disposals to include but not limited to income derived from the disposal and depreciated value, and validation of the asset disposed;

iv. Detail regarding the variables (dates, costs, standard asset lives, remaining lives, etc.) used in the prescribed formula, including the source of information and level of confidence in the accuracy of the variables used;

v. An excel spreadsheet clearly identifying the calculation from which the updated current Regulatory Period Closing RAB is calculated; and

vi. Completed Roll Forward Template as prescribed by the Commission, specifically the template for the Closing RAB.

f. Include a section detailing the Non-Network Assets proposed to be included in the roll forward RAB value and present the Independent Expert's review of the reasonableness of these assets including sufficient evidence that there exists no excessive investment, asset stranding or over-pricing.

g. (For DUs with refurbished assets) Present the Expert's review of the engineering reports prepared by the DU in relation to the refurbishment undertaken on the assets, and provide an expert opinion on the reasonableness of the new standard life assigned to an asset after the refurbishment;

[11] Refers to the time when the assets started commercial operation.
[12] Refer to Section 4.2.

h. Include a section detailing assets remaining in service beyond their regulatory asset lives[13] and present the Expert's review of the reasonability of the residual values used in the calculation of the proposed roll forward RAB value;

i. Include a section presenting the Expert's review of assets that are to be excluded and included[14] in the proposed roll forward RAB value;

j. Include a copy of the asset register[15], developed following the reviews undertaken under items (d), (e), (f), (g), (h) and (i). A separate asset register shall be developed for sub-transmission assets[16]; and

k. Include a certification stating that:

   i. The Expert was responsible for preparing the report;

   ii. That the report was prepared in accordance with the Handbook;

   iii. That the review was undertaken independently, there exists no conflict of interest, and that the Expert must not be related within the fourth civil degree of consanguinity or affinity to any of the directors, trustees, officers or their spouses of the DU;

   iv. That the Expert is not a stockholder of the DU under review;

   v. That the network was optimized correctly in accordance with the Handbook; and

   vi. That the report represents a fair and accurate reflection of the value of the RAB, based on the methodology described in this Handbook.

All sections of the report should include adequate information to allow a reader of the report to:

a. Understand all the data and information presented in the report;

b. Understand the calculations and verify the accuracy of the calculations;

c. Understand all the assumptions used by the Expert; and

d. Understand the extent to which professional judgment was exercised and the effect of that judgment in deriving the prosed results.

2.4 REQUIRED INFORMATION FOR THE DIRECTORS' CERTIFICATE

The purpose of the Directors' Certificate is to impose obligations on the DUs to ensure that they produce true, accurate, and reliable information. This will also encourage DUs to implement proper internal systems and controls.

---

[13] As per Clause 4.8.12 of the RDWR, there will be no further depreciation on assets remaining in service beyond their regulatory life. Moreover, at such time that these assets are finally removed from service, these assets should not form part of the disposal calculation as presented in Section 4.2 of this Handbook.

[14] As per Clause 4.8.8 of the RDWR, the RAB must only include assets that are 1) necessary to meet customer requirements; 2) except in the case of spares, are in service; 3) in the case of spares, are in reasonable quantities; and 4) in the case of easements, are clearly documented as being owned by the DU.

[15] Refer to Section 3.1.2 for the required information in an asset register

[16] As per Clause 1.7 of the RDWR, the asset register for sub-transmission assets shall include information on whether the asset is originally owned by the DU or a transferred sub-transmission asset. Valuation of sub-transmission assets shall follow the process referred to in Clause 4.8.11 of the RDWR promulgated in December 2021.

The Directors' Certificate must be completed and signed by a minimum of two board of directors of the DU and shall include the following:

a. A statement accepting the Independent Expert's Report and the validity of the Expert's certificate;

b. A statement certifying that the directors have made all reasonable enquiry, and to the best of their knowledge and belief, that the Roll Forward Information and the independent reports comply with the Commission's requirements as presented in this Handbook;

c. A statement that the Roll Forward Information was prepared in accordance with necessary internal controls to ensure that all the information provided is free from material misstatement;

d. In the case of non-compliance to specific requirements which does not lead to material misstatement, a detailed description of the non-compliance;

e. A statement that the DU, its Board, Officers, employees, or stockholders will not, in any way, interfere or influence the performance of the tasks of the independent expert as provided under this handbook. That any violation of this oath of non-interference shall be a ground for the ERC to reject or disapprove the submission of the DU, without prejudice to the legal actions that the Commission may thereafter take; and

f. A statement certifying that the Roll Forward Information has been properly extracted from the DU's relevant system(s) with the most accurate asset information, and that the key inputs to the Roll Forward Information have been reconciled to the different source documents and systems of the DU.

2.5 REVIEW OF DU SUBMITTED INFORMATION AND REPORTS

The ERC will review the independent reports submitted by the DUs. The DU shall ensure that the reports include accurate, sufficient, and verifiable information to enable the reviewer of the reports to independently assess the accuracy, validity and robustness of the Roll Forward Information in accordance with the methodologies prescribed in this Handbook.

In the review of the information submitted by the DUs, the following will be undertaken:

a. A compliance review of the submitted independent reports and the Directors' Certificate with additional detailed reviews as listed below;

b. Subsequent to the confirmation of the compliance, a detailed review of the Independent Expert's Report will be performed to ensure prudency and quality of all information and to undertake necessary adjustments to the proposed roll forward values if deemed necessary;

c. A reasonability test: It is recognized that for most network components modern equivalent asset (MEA) values will already be reflected in the standard replacement costs as used in the previous RAB valuations. Even so, and due to the fact that DUs are not implementing a revaluation or roll forward at similar times, the ERC retains the right to revisit replacement costs used in the previously approved valuations as well as consider any Asset Additions to ensure cost efficiency;

d. A detailed review of the models developed by the Expert, as required in 2.3 (d) used to recalculate the Opening RAB for the previous Regulatory Period which includes a detailed review of all inputs to the models;

e. A detailed review of the models developed by the Expert, as required in 2.3 (e) used to calculate the Closing RAB for the current Regulatory Period which includes a detailed review of all inputs to the models;

f. A detailed review of the optimization performed including:

   i. The optimization process undertaken; and

   ii. proposed optimization adjustments, including all calculations, assumptions used, and the bases of the proposed adjustments.

g. A review of the quality of supply criteria used by the DU for planning purposes as it applies to optimization;

h. If applicable, a detailed review of the age profiles of the assets, including detailed review of the calculation of the remaining lives of the assets, with particular focus on assets where the remaining lives cannot be accurately determined because of the absence of actual installation dates. Such review will include, assessing the method and all assumptions used to determine such asset ages and the subsequent calculation of the remaining lives;

i. If applicable, a review of the DU's database system containing the asset register and the process being undertaken by the DUs to accurately update and maintain the system, including its capability to generate the required asset registers and other relevant reports;

j. Applicable to relevant assets and period, a detailed review of the accuracy of the DU's asset register. This will include (but not limited to) field inspections of randomly selected assets[17]; examination of system diagrams and single-line diagrams and the comparison of this with the information included in the asset register; analysis of evidences supporting the asset register; and verification of the asset age information provided; and

k. Efficiency assessment of cost and quantity of assets through benchmarking for example Non-Network asset associated cost and quantity.

[17] The sampling method to be applied by the Expert shall be based on a statistically valid sampling methodology, resulting in a confidence level not lower than 95% with a margin of error below the materiality threshold of 5%. The sampling will be applied to Asset Additions as well as, if deemed necessary the most material (materiality threshold is 5%) asset categories.

3. ROLL FORWARD VARIABLES

3.1 INTRODUCTION

This section defines the key inputs to the calculation of the RAB which includes the following:

a. Opening RAB[18];

b. Asset Additions;

c. Depreciation; and

d. Asset Disposals.

The above key inputs will be used to calculate the closing value of the RAB to be carried forward at the start of the next Regulatory Period. The calculation of the Closing RAB is presented in more detail in the next section.

3.1.1 Allowed Assets in the RAB

Assets that are allowed to be included in the RAB of a DU are assets that are used for the efficient provision of the following services:

- Regulated Distribution Services;
- Distribution Connection Services; and
- Regulated Retail Services.

Below is a table exploring the assets associated with the above services constituting typical assets that could be owned and/or managed by a DU and therefore allowed to form part of the RAB.

Table 2: Breakdown of RAB[19]

REGULATED DISTRIBUTION ASSETS	DISTRIBUTION CONNECTION ASSETS	REGULATED RETAIL SERVICES ASSETS
Assets for providing: • Electricity conveyance from bulk or embedded generation connection points to end-user connection points • Administration and general management	Assets for providing: • Capability at a connection point • Conveyance of electricity • Planning, installation, maintenance, augmentation and operation of	Assets for providing: • Billing and collection • Customer support • Collection of bad debt • Energy trading • Electricity sales (excluding

[18] The Opening RAB may be based on an adjusted Closing RAB for the previous Regulatory Period as discussed in Section 4.2.1

[19] "Position Paper: Regulatory Reset for the July 2022 to June 2026 Fifth Regulatory Period for the First Entry Group of Privately Owned Distribution Utilities subject to Performance Based Regulation", Energy Regulatory Commission, December 2021.

REGULATED DISTRIBUTION ASSETS	DISTRIBUTION CONNECTION ASSETS	REGULATED RETAIL SERVICES ASSETS
of the regulated distribution services part of the business of a DU • Distribution ancillary services • Distribution information technology • Street and area lighting • Regulated distribution system control, metering and telecommunications • Transferred sub-transmission assets	distribution connection assets	generation and transmission costs) • Consumer metering installation[20]

3.1.2 RAB Asset Register

DU shall maintain an accurate asset register broken down into asset types and shall include all relevant information about the assets which will include, but will not be limited to the following:

- The different asset properties relevant to determining the value of an asset e.g. material type, size, capacity, ratings, installation and configuration details, and more;
- Original date of acquisition;
- Location of the asset and environment in which they are installed;
- Date of initial installation;
- Original estimate of asset life;
- Remaining asset life;

[20] As per Section 5.1.3 of the Position Paper, for purposes of the RAB, all metering equipment used for customers making up the captive market shall be included as Regulated Retail Services assets. Metering equipment supplied for customers in the contestable market or those connected under the supplier of last resort provision will not form part of the RAB, even where this equipment is owned by the DU as the recovery of these metering charges should form part of the metering charges negotiated with customers in these classes.

- Refurbishment record of an asset including corresponding extension of asset life due to refurbishment;
- Initial historical cost of the installed and commissioned asset, including all costs capitalized against the asset;
- Original purchase cost;
- Receipts or Proof of acquisitions in electronic format;
- All costs capitalized against the asset after installation, such as for refurbishment;
- Replacement cost;
- Unique asset identifier (to allow an asset to be identified in the field); and
- Asset utilization.

The DUs shall be responsible in maintaining a reliable and secure database system to store and maintain all relevant and accurate asset information with the ability to extract relevant reports and an asset register. Furthermore, it is expected that the asset registers will be disaggregated to allow individual assets to be identified and verified.

While the asset registers shall be detailed and comprehensive to simplify the calculation of the RAB through the roll forward approach, the different asset types in the DU's asset registers shall be aggregated into Asset Categories. The Roll Forward Template is presented in Appendix A of this Handbook.

3.1.3 Asset Categories

Consistent with the RDWR, the following is the required asset categorization for assets that will be included in the RAB:

Regulated Distribution Services Assets

i. Distribution services

   a. Land and Land Rights (dedicated to distribution purposes)

   b. Structures and Improvements (dedicated to distribution purposes)

   c. Substation Equipment

      - Power transformers
      - Switchgear
      - Protective equipment
      - Metering and control equipment
      - Communications equipment
      - Other station equipment

   d. Poles, Towers and Fixtures

   e. Overhead Conductors and Devices

f. Underground Conduits

g. Underground Conductors and Devices

h. Distribution Transformers

i. Power Conditioning Equipment

j. Meters, Metering Instruments & Metering Transformers (dedicated to distribution purposes)

k. Information technology equipment including cybersecurity (dedicated to distribution purposes)

l. Regulated Entity property on Consumers' Premises (not forming part of Distribution Connection Assets)

m. Street Lights and Signal Systems

n. Submarine Cables

ii. General Plant (Non-network Assets)

a. Land and Land Rights (non-network related)

b. Structures and Improvements (non-network related)

c. Office Furniture and Equipment

d. Transportation Equipment

e. Stores Equipment

f. Tools, Shop and Garage Equipment

g. Laboratory Equipment

h. Information systems equipment including cybersecurity (non-network related)

i. Power-operated Equipment

j. Communication Plant and Equipment

k. Miscellaneous Equipment

iii. Materials and Supplies, including spares[21]

iv. Transferred Sub-transmission Assets

v. Innovation

Distribution Connection Services Assets

i. Distribution services

[21] Materials and supplies, including spares, may be included in the RAB provided that the quantities are reasonable considering the historical reliability of the equipment and the quantity of items installed in the Distribution System.

a. Poles, Towers and Fixtures

b. Overhead Conductors and Devices

c. Underground Conduits

d. Underground Conductors and Devices

e. Distribution Transformers

f. Information technology equipment including cybersecurity (dedicated to Distribution Connection Services)

g. Streetlights and Signal Systems

ii. General Plant (Non-network Assets)

a. Land and Land Rights (non-network related)

b. Structures and Improvements (non-network related)

c. Office Furniture and Equipment

d. Transportation Equipment

e. Stores Equipment

f. Tools, Shop and Garage Equipment

g. Laboratory Equipment

h. Information systems equipment including cybersecurity (non-network related)

i. Power-operated Equipment

j. Communication Plant and Equipment

k. Miscellaneous Equipment

iii. Materials and Supplies, including spares[22]

Regulated Retail Services Assets

a. Meters, Metering Instruments & Metering Transformers – Consumer Import and Export metering

b. Land and Land Rights

c. Structures and Improvements

d. Office Furniture and Equipment

e. Transportation Equipment

[22] Materials and supplies, including spares, may be included in the RAB provided that the quantities are reasonable considering the historical reliability of the equipment, principles prudent acquisition and management of funds, adequate standardization of asset type timeliness of acquisition, and the quantity of items installed in the Distribution System.

f. Stores Equipment

g. Tools, Shop and Garage Equipment

h. Laboratory Equipment

i. Information systems equipment including cybersecurity

j. Communication Plant and Equipment

k. Miscellaneous Equipment

Any other Asset Categories specified by the ERC such as Assets for Regulatory and Governmental Requirements Compliance.

.

3.1.4 Description of Asset Categories and Asset Classes

The definition of each network Asset Category and the network Asset Classes that shall fall under each Asset Category are presented below.

Asset Category: Substation Equipment		
	Asset Class	**Description**
	Power Transformer	Grouped by voltage level and power ratings. Shall include foundations, conductors between high voltage line and the transformer terminals, high voltage, low voltage and control cable terminations at the transformer, and transformer earthing.
	Switchgear	Broken down into circuit breakers and disconnectors. Circuit breakers shall be grouped by type, voltage and size and shall include foundations, control cable terminations and control, metering and protections tests. Disconnectors shall be grouped based on voltage and rating.
	Protective Equipment	Shall be divided into current transformers, potential transformers, lightning arresters and protection schemes. Shall include all foundation, stands, conductor to the main circuit and all cable and conductor terminations. Shall also include metering potential transformers and current transformers.
	Metering and Control Equipment	Shall include all substation metering equipment that is used for power quality and

		revenue metering. Metering boxes and control panels shall be included. Control panels shall include all equipment for the operation of circuit breakers from the control room and shall include IED devices for monitoring circuit loads and voltages.
	Communications Equipment	Shall include all supervisory control and data acquisition (SCADA) and communications equipment.
	Other Station Equipment	Shall be divided into batteries and chargers, structures and busbars, and miscellaneous substation equipment, which consists of any substation item that cannot be associated with any particular piece of equipment, such as substation earthing. Batteries and chargers shall be sub-categorized based on capacity and ratings, while structures and busbars shall be based on voltage and configuration.
Asset Category: Poles, Towers and Fixtures		
	Asset Class	**Definition**
	Pole	Shall be sub-divided based on construction material and height. Shall include all line hardware required for installing the pole (e.g. foundations, guys, and more).
	Pole Top	Shall include cross-arms, insulators, braces, clamps, nuts and bolts to secure the cross-arms, and insulators to the pole. Shall also include any material required to secure the conductors to the insulators
Asset Category: Overhead Conductors and Devices		
	Asset Class	**Definition**
	Overhead Conductor	Shall be further divided based on conductor material and cross-sectional area.
	Overhead Line Devices	Shall include fuse cut-outs, surge arresters, line switches, line sectionalizers and reclosers.
Asset Category: Underground Conduits		
Asset Category: Underground Conductors and Devices		

	Asset Class	Definition
	Underground Conductor	Shall be based on conductor type. Shall include cable joints and terminations along with all trenching and reinstatement.
	Underground Devices	Shall include ground-mounted switching devices, further sub-divided based on the voltage rating. Shall include foundations, cable terminations and earthing.
Asset Category: Distribution Transformer		
	Shall be sub-categorized based on voltage phase and rating. Shall include the brackets for mounting on poles (or foundations if ground mounted), the conductor between the fuse unit and the transformer terminals, all connectors and terminations and the transformer earthing.	
Asset Category: Power Conditioning Equipment		
	Shall include capacitors banks for power factor correction, voltage regulators, and generators used for spinning reserve or voltage stability, and VAR compensators[23]. Capacitors shall be further subcategorized as substation capacitors and line capacitors. Substation capacitors are installed in sub-transmission and distribution substations and generally of a larger capacitor. Capacitors shall include all foundations, structures, earthing, switching reactors and potential transformers. Line capacitors are installed on distribution circuits and includes all control equipment (if switched), brackets for installation on the poles, and conductors required shall be included in this subcategory.	
Asset Category: Meters, Instruments and Metering Transformers		
	Shall include meter, instrument transformers (if any), metering cabinets, sockets and seals.	
Asset Category: Street Light and Signal Systems		
	Shall be further sub-divided according to voltage for cables and wattage for street lights. Each standalone streetlight pole shall consist of a pole, an outreach arm and fitting complete with luminaire. Each street light installed on existing power poles shall consist only of an outreach arm and fitting with luminaire.	
Asset Category: Submarine Cables		
Asset Category: Overhead Conductors and Devices (distribution connection services) – Service Drops		

[23] Power conditioning equipment as defined under Section 4.8.6 (a) of the: "Revised Rules for Setting Distribution Wheeling Rates for Privately Owned Electricity Distribution Utilities Operating Under Performance Based Regulation", Energy Regulatory Commission, December 2021.

	Shall be grouped based on cross sectional area. Shall include termination for the cable at the transformer and the customer premises.

3.2 OPENING RAB FOR REGULATORY YEAR 1

The Opening RAB would be the latest approved Opening RAB of the DU.[24] That is, in the case of the DUs who have undergone valuation in the previous Regulatory Period, the last approved Opening RAB would be the opening value to be rolled forward.

The Opening RAB shall be broken down into Asset Categories as described in Section 3.1.3.

3.3 ASSET ADDITIONS FOR CURRENT REGULATORY PERIOD[25]

Asset Additions for the current Regulatory Period refer to the actual nominal value of Asset Additions for each year of the current Regulatory Period. Forecast Asset Additions will be required as inputs for the final year(s) of the current Regulatory Period in the event that the Closing RAB value is assessed prior to the end of the current Regulatory Period.

Categories of Asset Additions should match the Asset Categories in the Opening RAB in order to allow for consistent depreciation across asset classifications.

3.4 ASSET DISPOSALS FOR CURRENT REGULATORY PERIOD[26]

Asset Disposals for the current Regulatory Period[27] refer to the actual nominal Asset Disposals that have taken place in each regulatory year of the current Regulatory Period.

The value of Asset Disposals shall be calculated as the difference between the actual income derived from disposing these assets (after accounting for any costs associated with the disposal) and the depreciated value of assets disposed.

Forecast Asset Disposals will be required as inputs for the final year(s) of the current Regulatory Period.

Categories of Asset Disposals should match the Asset Categories in the Opening RAB.

3.5 DEPRECIATION

To calculate the actual nominal depreciation, the straight-line method of depreciation shall be adopted, consistent with the method used for previous Regulatory Periods. Depreciation shall be applied where an existing asset's remaining life is less than the asset's standard life.

---

[24] Please refer to Section 4.2.1 whereby the approved Opening RAB could be required to be adjusted due to the actual additions, disposals and depreciation during the latter part of the previous Regulatory Period and/or the reasonableness test adjustments of the ERC.

[25] The period under review may extend beyond the current Regulatory Period, as deemed necessary and as determined by the ERC.

[26] The period under review may extend beyond the current Regulatory Period, as deemed necessary and as determined by the ERC.

[27] The period under review may extend beyond the current Regulatory Period, as deemed necessary and as determined by the ERC.

Yearly depreciation shall be calculated for the Opening RAB as well as for the Asset Additions.

To calculate the depreciation, the standard asset lives and remaining lives for all assets (except land and land rights) need to be established, as can be viewed below.

3.5.1 Standard Asset Lives[28]

The standard asset lives for each network Asset Class is as follows:

Table 3: Standard Asset Lives for each Network Asset Class

Asset Category: Substation Equipment			
	Asset Class		**Standard Asset Life**
	Power Transformer		40
	Asset Class	**Asset Sub-Class**	**Standard Asset Life**
	Switchgear	Circuit Breakers	30
		Disconnectors	30
	Asset Class	**Asset Sub-Class**	**Standard Asset Life**
	Protective Equipment	Current Transformers	30
		Potential Transformers	30
		Lightning Arresters	15
		Protection Schemes	30 (Mechanical), 15 (Electronic)
	Asset Class		**Standard Asset Life**
	Metering and Control Equipment		30
	Asset Class		**Standard Asset Life**
	Communications Equipment		10
	Asset Class	**Asset Sub-Class**	**Standard Asset Life**
	Other Station Equipment	Batteries and Chargers	10
		Structures and Busbars	The asset life of structures and busbars shall be a weighted life (by value) of the major components in the module i.e. circuit breaker,

[28] For purposes of the roll forward, and due to the aggregated nature of asset categories in the roll forward process and in the Roll Forward Template, the standard asset lives per Asset Category shall be calculated as the weighted average asset lives of the standard asset lives as depicted in this section.

			disconnectors and structures, excluding power transformers
		Miscellaneous Substation Equipment	The asset life of miscellaneous substation equipment shall be a weighted life (by value) of the major components in the module i.e. circuit breaker, disconnectors and structures, excluding power transformers
Asset Category: Poles, Towers and Fixtures			
	Asset Class	**Asset Sub-Class**	**Standard Asset Life**
	Pole	Wood	20
		Concrete	30
		Steel	40
		Steel Tower (Sub-transmission)	50
		Galvanized Iron	30
	Asset Class		**Standard Asset Life**
	Pole Top		30
Asset Category: Overhead Conductors and Devices			
	Asset Class		**Standard Asset Life**
	Overhead Conductor		30
	Asset Class	**Asset Sub-Class**	**Standard Asset Life**
	Overhead Line Devices	Fuse Cut-out	30
		Surge Arresters	30
		Line Switches, Circuit Breakers and Disconnectors (Sectionalizers and Disconnectors)	30
		Reclosers	30
Asset Category: Underground Conduits **Asset Category: Underground Conductors and Devices**			
	Asset Class		**Standard Asset Life**

	Underground Conductor		35
	Underground Devices		35
Asset Category: Distribution Transformer			
Asset Category			**Standard Asset Life**
Distribution Transformer			30
Asset Category: Power Conditioning Equipment			
	Asset Class	**Asset Sub-Class**	**Standard Asset Life**
	Capacitor	Substation Capacitor	35
		Line Capacitor	30
	Asset Class		**Standard Asset Life**
	Voltage Regulator		35
Asset Category: Meters, Instruments and Metering Transformers			
	Asset Class		**Standard Asset Life**
	Meters		25 (ElectroMechanical), 15 (Electronic)
	Current Transformers		25
	Potential Transformers		25
Asset Category: Street Light and Signal System			
Asset Category			**Standard Asset Life**
Street Light and Signal System			30
Asset Category: Submarine Cables			
Asset Category: Overhead Conductors and Devices (distribution connection services) – Service Drops			
Asset Category			**Standard Asset Life**
Service Drops			30

Standard asset lives of Network Assets not provided in Table 3 above shall be established on a comparable basis with the standard asset lives provided in the aforementioned table, but shall not exceed such standard asset lives for comparable assets. Furthermore, the assessment of the standard asset lives of Network Assets not provided in Table 3 shall include the examination

of service records, physical inspection of the assets, and discussion with maintenance personnel.

Standard asset lives used in the determination of the Closing RAB for the previous Regulatory Period (up to the First Closing Date[29]) will be used in the deprecation calculation of all assets forming part of the Closing RAB for the previous Regulatory Period (up to the First Closing Date). For assets added after the First Closing Date, the standard asset lives presented in Table 3 above or any updated list of standard asset lives issued by the ERC after the adoption of this Handbook shall be used. If there are reasons to update the standard lives, the DUs may include it as part of their application/rolled-forward RAB report with sufficient justifications subject to review of the ERC.

Standard asset lives for Non-Network assets shall be based on Philippine Accounting Standard (PAS) 16, which corresponds to International Accounting Standard (IAS) 16 – Property, Plant and Equipment. PAS shall also be the primary accounting standard governing the valuation of non-network assets.

3.5.2 Asset Lives for Refurbished Assets

The DU shall be responsible for assigning an asset life to a refurbished asset. The asset life shall be effective from the time of refurbishment and shall be for a period not greater than the standard asset life of such asset. The DU shall be required to prepare an engineering report presenting the details of the refurbishment work undertaken and justifying the basis of the new assigned asset life. These reports, if any, should form part of the Independent Expert's Report.[30]

3.5.3 Remaining Lives[31]

The life of an asset commences when it is first installed/commissioned/refurbished. The remaining lives of assets are then determined by subtracting the age of the assets from their standard asset lives. The age of an asset shall be determined from the DUs records of installation dates establishing the age.

For assets with no record of installation/commissioning dates, the DU shall provide sufficient evidence for the basis of the assigned remaining life of the asset, which shall be consequently reviewed in detail by the Commission as provided for in Section 2.6.

3.5.4 Residual Values

The residual value of an asset must be estimated to perform a depreciation calculation in cases when the asset remains in service beyond the standard asset life; alternatively, such asset may be sold to a secondary user or for salvage value.

In principle when an asset reaches the end of its standard asset life, it has zero value under the straight-line depreciation method. Specific assets, however, may remain in service beyond this period.

---

[29] First Closing Date is defined under Section 1.5 of this Handbook.

[30] Please refer to Section 2.

[31] For purposes of the roll forward, and due to the aggregated nature of asset categories in the roll forward process and in the Roll Forward Template, the remaining lives per Asset Category shall be calculated as the weighted average remaining asset lives of all assets in the relevant Asset Category.

In the past, the adopted approach was to allocate a residual value of 5%[32] of the optimized replacement cost of assets used beyond their standard asset lives to recognize their economic value[33] to the network and provide an incentive for DUs to extend the useful lives of assets. Going forward, the ERC will no longer automatically adopt a 5% residual value and will consider the residual value justifications submitted as part of the requirements under Section 2.3.h of this Handbook will be considered by the Commission. Note that there will be no further depreciation of the asset once it reaches the end of its standard asset life.

3.6 EXCLUDED VARIABLES

Consistent with previous Regulatory Periods, value added tax (VAT) shall be excluded from the RAB calculation. Inflationary factors shall also be excluded from the RAB calculation at the stage of applying the roll forward as stipulated in this Handbook.

[32] 1. COA Circular No. 2017-004 dated December 13, 2017, Guidelines on the preparation of financial statements and other financial reports and implementation of the Philippine Financial Reporting Standards by Government Corporations Classified as Government Business enterprises and Philippine Public Sector Accounting Standards by Non-Government Business Enterprises.
2. Government Accounting Manual Volume 1, Chapter 10, Property Plant and Equipment, Section 27, Depreciation under paragraph i and j.

[33] Value placed on an asset based on the benefit that could be derived from the asset.

4. ROLL FORWARD PROCEDURES

4.1 INTRODUCTION

The objective of the roll forward procedure described in this section is to determine the Closing RAB for each DU for each Regulatory Period. This Closing RAB value becomes the Opening RAB value to make a regulatory determination for the next Regulatory Period.

The roll-forward procedure consists of three main steps. The first is the removal of any penalty or benefit on returns associated with differences between the actual and forecast Asset Additions, Asset Disposals and depreciation associated with the end period of the previous Regulatory Period[34].

Secondly, the final Closing RAB is determined by calculating the actual nominal Closing RAB for each regulatory year of the current Regulatory Period. This yearly calculation will utilize actual Asset Additions, Asset Disposals and depreciation during the current Regulatory Period.

Lastly, the procedure also consists of the application of network optimization.

4.2 ROLL FORWARD PROCEDURE

This subsection provides the procedure to establish the nominal Closing RAB for each year of the subject Regulatory Period. The Closing RAB of the final year of the subject Regulatory Period will also be the Opening RAB value for the next Regulatory Period.

For this procedure, work in progress will not form part of Asset Additions and will only appear in the year of commissioning (energization).

This procedure assumes the occurrence of actual Asset Additions and Asset Disposals during the subject Regulatory Period. Where the procedure is applied prior to the end of the current Regulatory period it is allowed that the Asset Additions and Asset Disposals be based on forecast values for the final year(s) of the Regulatory Period. Adjustments if any in such forecast values covering the end of the current Regulatory Period will be done using actual asset additions and asset disposals in the next Regulatory Period as per the procedure describe in Section 4.2.1.

Variances from the Allowed Capital Expenditure may be realized at the start of the subject Regulatory Period. These variances from the allowed CAPEX may be caused by unforeseen increases or decreases in demand, actual costs or even errors in forecasting at the onset. These potential variances do not affect the outcome of the roll forward procedure as it is based on actual Asset Additions with adjustment as described in the following section.

The formula for calculating the Closing RAB for each year of the subject Regulatory Period[35]:

Opening RAB for regulatory year 1

[34] As an example, for DUs in the Third Regulatory Period, the Previous Regulatory Period refers to the Second Regulatory Period.
[35] The period under review may extend beyond the current Regulatory Period, as deemed necessary and as determined by the ERC.

+ Actual nominal Asset Additions for regulatory year 1

– Actual nominal regulatory depreciation for regulatory year 1

– Actual nominal Asset Disposals for regulatory year 1

= Closing RAB for regulatory year 1 = Opening RAB for regulatory year 2

+ Actual nominal Asset Additions for regulatory year 2

– Actual nominal regulatory depreciation for regulatory year 2

– Actual nominal Asset Disposals for regulatory year 2

= Closing RAB for regulatory year 2 = Opening RAB for regulatory year 3

(Additional Years)

= Closing RAB for regulatory year X[36]

+ - Optimization[37]

= Closing RAB for the current Regulatory Period

Where:

Opening RAB for regulatory year 1 = closing RAB for the final regulatory year of the previous Regulatory Period, which was calculated following Section 4.2.1

Actual nominal Asset Additions and Asset Disposals are assumed to be spread evenly over a regulatory year and therefore the reported values are assumed to be in mid-year terms.

For clarity, actual nominal regulatory depreciation includes depreciation on the new Opening RAB as well as depreciation on the previous year's actual additions.

4.2.1 Adjusting Asset Additions, Asset Disposals and Depreciation for the Previous Regulatory Period

This subsection is only relevant when the Closing RAB of the previous Regulatory Period was established by utilizing forecast Asset Additions, forecast Asset Disposals or forecast depreciation values for part or all of the previous Regulatory Period. This utilization of forecast values at the end of the Regulatory Period is expected because the Closing RAB for a Regulatory Period is generally calculated before the end of the said period. If the Closing RAB for the previous Regulatory Period were based on actual Asset Additions, actual Asset Disposals, and actual depreciation, the procedure described in this subsection would not be required.

[36] X depicts the final year of the current Regulatory Period.

[37] Optimization in the formula should include newly identified optimizations apart from the optimization approved in the Opening RAB for regulatory year 1 and is assumed that it may include the following: (1) assets determined to form part of the RAB which were previously optimized from the Opening RAB for regulatory year 1 and the depreciated value will be added to the RAB; and (2) assets included in the RAB but will now be optimized and that the depreciated value of which will be subtracted from the RAB. For the avoidance of doubt, optimizations in the Opening RAB for regulatory year 1 that should remain optimized will have no effect on the RAB.

If the Closing RAB of the previous period was based on forecast values, the following procedure should be followed to adjust such values, which will be applied during the procedure for calculating the Closing RAB for the Regulatory Period under review.

For this procedure, work in progress will not form part of Asset Additions, and such assets will only appear in the year of commissioning (energization).

The formula for calculating the Opening RAB for the current Regulatory Period:

Opening RAB for the final year of the previous Regulatory Period

+ actual nominal Asset Additions

– actual nominal Disposals

– actual nominal regulatory depreciation

= Opening RAB for the first regulatory year of the current Regulatory Period

Where:

Actual disposals are defined as the Asset Disposal value minus the value obtained from the asset's disposal.

4.3 OPTIMIZATION

4.3.1 Introduction

The purpose of optimization is to ensure that the RAB reflects an efficient set of assets that would achieve the required level of service. An "incremental optimization" approach is to be followed in which the existing network is examined and changes made to ratings, configurations, designs or materials to optimize the network configuration and its component asset having regard to such issues as excess capacity, redundancy and over-design. DUs shall adopt standard sizes/ratings based on least-cost considerations and optimization also takes this practice into account. The optimization to be undertaken will not be concerned with improving the system from its current state as optimization cannot result in an increased network system asset replacement cost.

Optimization consists of five stages:

i. removal of stranded assets in the RAB;

ii. optimization of the DU's network configuration;

iii. optimization of the capacity/voltage of elements in the DU's network;

iv. optimization of network engineering; and

v. optimization of materials and supplies, including spares.

4.3.2 Optimization Constraints

The optimization shall be based on a reasonably expected level of use of the asset, which is determined by reference to the required level of service potential or output consistent with both the reasonably foreseeable future demand and the objective of minimizing the whole of life costs.

The optimization process shall be carried out subject to the following constraints:

i. the potential level of service of the optimized network shall not exceed that of the existing network, and the performance of any part of the optimized network shall not exceed the DU's standard quality of supply criteria, as disclosed in the Independent Expert's Report. If non-standard contracts with individual customers exist that require the DU to provide an enhanced quality of supply, these assets[38], shall be optimized out of the RAB on the basis that these assets are required by the individual customer rather than the DU and that customer-specific rates (falling outside normal distribution wheeling rates) will apply to such installations;

ii. the location of points of connection to other networks or transmission grid exit points shall be assumed to be fixed. However, where a fixed point of connection can be readily bypassed and replaced with a more cost-efficient network arrangement, then that fixed point of connection shall be deleted for valuation purposes and the more cost-efficient network included;

iii. the location and number of connection points to consumers shall be assumed fixed;

iv. the optimized network should only use the voltage levels used on the existing network. This does not preclude existing equipment from being optimized down to a lower standard network voltage, but there is no requirement to optimize down to a non-standard voltage level; and

v. the optimized network shall use standard equipment/fitting ratings/sizes to optimize construction/maintenance practices. This does not preclude oversized existing equipment being optimized down to standard sizes used by the DU or typical sizes used by other DU's. However, there is no requirement to optimize equipment sizes down to non-standard equipment that is not readily available in the Philippines.

4.3.3 Quality of Supply

The optimized network shall be designed to supply the allowed future load growth. The quality of supply of the optimized network shall match the current level, except where this is greater than the disclosed quality of supply criteria.

The DU's quality of supply criteria currently used as a basis for network design shall be disclosed in the Independent Expert's Report. The Quality of Supply criteria shall be based on:

i. the DU's analysis of customer requirements;

ii. the DU's assessment of network maintenance requirements and costs;

iii. the standards and practices that have historically been used to develop the existing sub-transmission and distribution networks; and

iv. the DUs assessment of widely accepted best practice followed by other effective DUs in the Philippines and internationally, for similar distribution networks and customer types.

[38] Refers to assets used to enhance the quality of supply.

Relevant quality of supply criteria include:

i. the degree of security (redundancy) in different circumstances or localities;

ii. target reliability indices for different areas of the network. For example, central business, urban and rural districts;

iii. voltage regulation criteria; and

iv. levels of electrical losses.

The degree of security may be disclosed either in probabilistic or deterministic terms. A deterministic approach could reference the level of in-built redundancy, for example, as (n) or (n-1) or (n-2) component redundancy. An (n) security level implies no component redundancy so that if a component fails, then consumer supply is lost. An (n-1) security level implies that consumer supply is not interrupted in the event of any single component outage[39]. A less reliable (n-1 switched) security level involves the loss of consumer supply after a single component outage, but with subsequent consumer supply restoration via network switching. Irrespective of whether probabilistic or deterministic criteria are used, it is necessary for a DU to express its degree of security criteria so that the optimization process is transparent and can be shown to have been applied consistently across all sections of the network.

Existing Distribution System assets that provide a quality of supply greater than that disclosed by the DU shall be optimized out. As mentioned earlier, improved quality of supply is allowed as part of non-standard contracts but are not to be included in the RAB.

4.3.4 Future Load Growth

The maximum capacity of any part of a DU's optimized network shall be determined by considering the projected future load growth for the relevant planning period in tandem with the DU's quality of supply criteria. However, in no case shall the optimized network capacity exceed the existing capacity.

To ensure compliance, the DU's existing peak demand and forecast electrical demand levels that are used as a basis for optimization shall be disclosed in the Independent Expert's Report. The forecast electrical demand values should account for any potential to reduce peak demand by applying of cost-effective demand-side management practices. As a minimum, the existing and forecast electrical demand levels shall be provided for each (i) point of supply, (ii) distribution substation, (iii) sub-transmission feeder and (iv) primary distribution feeder. Clear justification and a detailed derivation of the load growth forecasts are required. Both the existing maximum demand and the highest forecast maximum demand during the planning period shall be provided. Allowances should be made, where possible, for different growth rates in different parts of the network. Existing electrical demands, power factors or diversity factors may be estimated where metering or SCADA data is not available. Estimated quantities should be clearly identified in the DU's disclosed forecast electrical demand[40].

The network demand forecast should:

[39] A completely uninterrupted supply in the event of a component failure is mostly not feasible and, depending on the definition adopted, supply restoration within 1 minute, 3 minutes or 5 minutes are usually considered to be no-break.

[40] The demand forecast figures presented by DUs should be consistent with those used to forecast the basis of its capital expenditure on network growth, for the next Regulatory Period.

i. be based on peak kilo-volt-ampere (kVA) values and include an indication of the relevant electrical power factors;

ii. clearly outline any diversity factors that may exist in relation to the peak loads on (i) the entire network, (ii) the zone substations and (iii) the primary distribution feeders;

iii. be based on "system normal conditions" and only consider the peak electrical demand on each of the relevant network components without load transfer due to network contingencies; and

iv. include only future electrical loads that can reasonably be expected to be supplied from the DU's network. Any loads that are forecast to emerge outside of a DU's franchise area should be noted explicitly in the Independent Expert's Report.

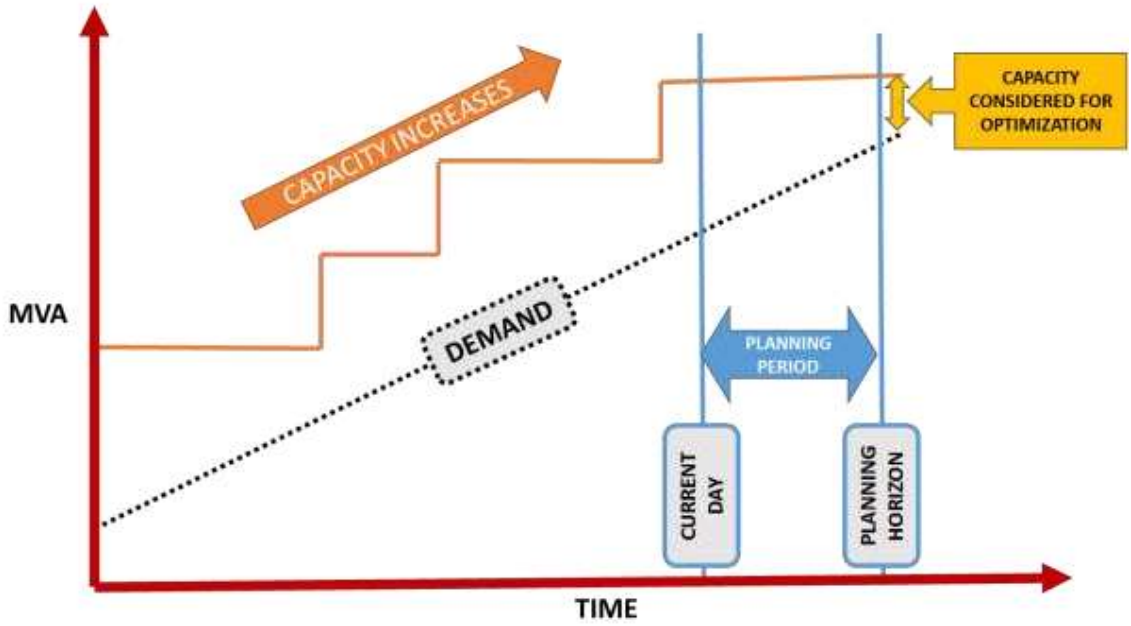
The Independent Expert's Report shall also present any separately identifiable new load or load increment that is forecasted to be (i) 5% of the DU's existing maximum demand or (ii) 5 megawatts (MW) (whichever is the lower).

The planning periods over which future load growth can be allowed for shall not exceed those outlined in the following table:

Table 4: Optimization Planning Horizon

NETWORK COMPONENTS	PLANNING HORIZON
Points of connection to the transmission network Delivery Point Substation Sub-transmission (HV) lines/cables Distribution Substations (excluding transformers)	15 years
Distribution Substation Transformers Primary Distribution (MV) lines/cables	10 years
Distribution/Line Transformers Secondary Distribution (LV) lines/cables Other Distribution Assets	5 years

The following figure illustrates the progression of a network asset's capacity and electrical demand graphically. It also shows the extent of the spare capacity at the end of the planning period, which would be subject to potential optimization under this Handbook's optimization methodology.



4.3.5 The Process of Optimization

Optimization of the network shall be undertaken on a systematic basis. The optimization process must examine the existing network and determine whether a more cost-efficient design could meet the DU's quality of supply criteria throughout the allowed planning period. Optimization shall be undertaken systematically across the network and shall include, in particular, the following network components: i. points of supply; ii. delivery point substations; iii. sub-transmission lines and distribution substations; iv. primary distribution switching stations; v. the primary distribution circuits/feeders (including distribution transformers); and vi. the secondary (low voltage) distribution system.

Removal of Stranded Assets in the RAB

Any asset not required to supply distribution services to the DU's existing customers, and which could therefore be disconnected, shall be identified and excluded from the optimized network.

Optimization of the DU's Network Configuration

Optimization of the system configuration shall be carried out by considering alternative configurations subject to the constraints on optimization and following the relevant quality of supply criteria. The optimized configuration is the one that supplies the forecast load at the end of the allowed planning period and meets the disclosed quality of supply requirements in the most cost-efficient manner.

In the process of optimizing the system configuration, specific assets or groups of assets may become excess to requirements. They should be assigned a zero value, while other new assets may need to be notionally brought in. The minimum optimization tests to be carried out by DUs in optimizing the system configuration are set out in Appendix B.

Optimization of the Capacity/Voltage of Elements in the DU's Network

After the system's configuration has been optimized, the elements within that system shall be optimized by considering whether lower capacity/voltage, more cost-efficient elements would be

adequate. The minimum tests to be carried out by DUs in optimizing the network capacity are set out in Appendix B.

Civil engineering works such as spare ducts, control buildings, cable tunnels and switchyard bays not currently used shall be optimized out unless they will be required to meet the allowed future load growth within a reasonable growth period, coupled with meeting the disclosed quality of supply criteria.

Optimization of Network Engineering

As part of optimizing the network, the engineering of the network shall be examined to confirm that the optimized asset base is not over-engineered, given the disclosed quality of supply criteria. Over-engineering may occur if parts of the existing asset base are engineered to a standard that exceeds the DU's current practice or if a more cost-efficient engineering arrangement or configuration would be used if the existing assets were replaced. The DU's design and construction standards, and the standard of engineering applied to its most recent projects should be used as the benchmark for this test.

Where a more cost-efficient arrangement would result if the required level of service were provided by applying the DU's existing engineering standards, the relevant assets shall be replaced by a notional asset arrangement that reflects current best practice. The minimum tests to be carried out by DUs in optimizing the network engineering are set out in Appendix B.

Optimization of Materials and Supplies, Including Spares

Network materials and supplies, including spares may be included in the RAB as long as the spares are suitable replacements for the assets installed in the network. However, the number of spares in the RAB shall not exceed the reasonable quantity of spares required to meet the DU's disclosed quality of supply criteria.

Stranded assets may be valued as network spares, subject to the criteria above. Stranded assets not required as network spares shall be assigned a zero value to calculate the RAB.

Determining the Optimized Replacement Cost

Once the optimized system has been determined, those parts of the optimized network that are different from the existing network shall be revalued. This entails applying the replacement costs of MEAs to the optimized notional network. A schedule of all network optimizations and details of the valuation impact of each optimization, including details of the assets removed as stranded assets, shall be included in the Independent Expert's Report.

When assets are notionally brought into the network due to the optimization process, they should be valued at the replacement costs of MEA to determine the total replacement cost.

5. LIST OF ACRONYMS AND SHORTCUTS

ACRONYM AND SHORTCUT	TERM
ARR	Annual Revenue Requirement
CAPEX	Capital Expenditure
DRC	Depreciated Replacement Cost
DU	Privately Owned Electricity Distribution Utility
ERC/Commission	Energy Regulatory Commission
Handbook	Roll Forward Handbook
HV	High Voltage
kV	Kilovolt
kVA	Kilo-volt-ampere
LV	Low Voltage
MEA	Modern Equivalent Asset
MV	Medium Voltage
MVA	Mega-volt-ampere
MW	Megawatt
OLTC	Off Load Tap Changer
PBR	Performance Based Regulation
POS	Points of Supply
RAB	Regulatory Asset Base
RDWR	Rules for Setting Distribution Wheeling Rates
SCADA	Supervisory Control and Data Acquisition
VAT	Value Added Tax

APPENDIX A: ROLL FORWARD TEMPLATE SAMPLES

All values must be provided as *Thousands*
Please complete areas identified as

Asset Categories	Opening Asset Value (Approved Closing RAB for previous Regulatory Period)	Weighted Average Remaining Life	Weighted Average Standard Asset Life	Actual Nominal Capex Final Year	Forecast Nominal Capex Final Year
Regulated Distribution Services Assets					
1. Distribution Services					
a. Land and Land Rights (dedicated to distribution purposes)					
b. Structures and Improvements (dedicated to distribution purposes)					
c. Substation Equipment					
- Power Transformers					
- Switchgear					
- Protective Equipment					
- Metering and Control Equipment					
- Communications Equipment					
- Other Station Equipment					
d. Poles, Towers and Fixtures					
e. Overhead Conductors and Devices					
f. Underground Conduits					
g. Underground Conductors and Devices					
h. Distribution Transformers					
i. Power Conditioning Equipment					
j. Meters, Metering Instruments & Metering Transformers (dedicated to distribution purposes)					
k. Information Technology Equipment (dedicated to distribution purposes)					
l. Regulated Entity Property on Consumers' Premises (not forming part of Distribution Connection Assets)					
m. Street Lights and Signal Systems					
n. Submarine Cables					
2. General Plant (Non-network Assets)					
a. Land and Land Rights (non-network related)					
b. Structures and Improvements (non-network related)					
c. Office Furniture and Equipment					
d. Transportation Equipment					
e. Stores Equipment					
f. Tools, Shop and Garage Equipment					
g. Laboratory Equipment					
h. Information Systems Equipment (non-network related)					
i. Power-operated Equipment					
j. Communication Plant and Equipment					
k. Miscellaneous Equipment					
3. Materials and Supplines, including spares					
4. Transferred Sub-transmission Assets					
Distribution Connection Services Assets					
1. Distribution Services					
a. Poles, Towers and Fixtures					
b. Overhead Conductors and Devices					
c. Underground Conduits					
d. Underground Conductors and Devices					
e. Distribution Transformers					

All values must be provided as *Thousands*
Please complete areas identified as
Year 1 refers to the first year after the previous Regulatory Period

Asset Categories	Newly Adjusted Opening RAB for the first regulatory year of the current Regulatory Period *(refer Opening RAB sheet)*	Optimization Adjustment for current Regulatory Period	Year 1 Actual Nominal Asset Additions	Year 2 Actual Nominal Asset Additions	Year 3 Actual Nominal Asset Additions	Year 4 Actual Nominal Asset Additions
Regulated Distribution Services Assets						
1. Distribution Services						
a. Land and Land Rights (dedicated to distribution purposes)						
b. Structures and Improvements (dedicated to distribution purposes)						
c. Substation Equipment						
- Power Transformers						
- Switchgear						
- Protective Equipment						
- Metering and Control Equipment						
- Communications Equipment						
- Other Station Equipment						
d. Poles, Towers and Fixtures						
e. Overhead Conductors and Devices						
f. Underground Conduits						
g. Underground Conductors and Devices						
h. Distribution Transformers						
i. Power Conditioning Equipment						
j. Meters, Metering Instruments & Metering Transformers (dedicated to distribution purposes)						
k. Information Technology Equipment (dedicated to distribution purposes)						
l. Regulated Entity Property on Consumers' Premises (not forming part of Distribution Connection Assets)						
m. Street Lights and Signal Systems						
n. Submarine Cables						
2. General Plant (Non-network Assets)						
a. Land and Land Rights (non-network related)						
b. Structures and Improvements (non-network related)						
c. Office Furniture and Equipment						
d. Transportation Equipment						
e. Stores Equipment						
f. Tools, Shop and Garage Equipment						
g. Laboratory Equipment						
h. Information Systems Equipment (non-network related)						
i. Power-operated Equipment						
j. Communication Plant and Equipment						
k. Miscellaneous Equipment						
3. Materials and Supplines, including spares						
4. Transferred Sub-transmission Assets						
Distribution Connection Services Assets						
1. Distribution Services						
a. Poles, Towers and Fixtures						
b. Overhead Conductors and Devices						
c. Underground Conduits						
d. Underground Conductors and Devices						
e. Distribution Transformers						
f. Information Technology Equipment (dedicated to Distribution Connection Services)						
2. General Plant (Non-network Assets)						

APPENDIX B: OPTIMIZATION[41]

B.1 Introduction

The optimization of a DU's assets shall be undertaken in accordance with the requirements of Section 4. The general approach is to systematically test (i) the network configuration, (ii) the network capacity/voltage and (iii) the network engineering against the DU's disclosed quality of supply criteria. The network is adjusted in an incremental fashion in situations where it is found that a more cost-efficient design will meet the pre-determined criteria. This section describes the **minimum tests** that shall be applied in all cases in order to optimize a DU's network in accordance with the requirements of this Handbook.

Note that the intent is to arrive at the most cost-effective DU supply network that meets with (i) the DUs quality of supply and (ii) the forecast electrical demand for the allowed planning period, within the boundaries of the optimization method.

Optimization of the DU's Network Configuration

B.2 Points of Supply and Delivery Point Substations

Issue (a): Whether all the existing points of supply (POS) and delivery point substations are required, given the allowed quality of supply criteria.

Approach: The location and supply voltage for the existing POS and delivery point substations may be considered fixed. All POS shall be tested to determine whether a more cost-efficient DU network would result if the POS were eliminated and the load supplied from adjacent POS. If possible, the POS shall be optimized out and replaced with a notional and more cost-efficient DU network. For example, an existing DU network is supplied via 4 x 69kV lines emanating from two POS. The network loading levels are light, and there is little prospect of significant future load growth within the allowed planning periods outlined in Section 4.3.4. The existing and future electrical load can be comfortably supplied using a more economical arrangement, involving only 2 x 69kV lines emanating from one POS, which still meets with the DU's disclosed quality of supply criteria. The other 2 x 69kV lines (and any associated Delivery Point Substation termination equipment, land and easements owned by the DU) should be optimized out.

B.3 Sub-transmission (HV) Circuits

Issue (a): Whether the number of sub-transmission lines/cables exceeds the number required given the DU's quality of supply criteria and the forecast future load growth within the allowed planning period.

Approach: The route of each line/cable may be considered fixed. Assess the required number of sub-transmission lines/cables concerning the allowed quality of supply criteria and future load growth within the allowed planning period. Optimize out those lines/cables that are not required. For example, an existing DU's sub-transmission network is supplied from a single delivery point substation via 3 x 69kV lines. Given the projected loading levels (over the entire planning period), the existing DU network is afforded with (N-2) security, whilst the DU's quality of supply criteria indicates only (N-1) security is required. One of the 69kV lines, together with any associated delivery point

[41] "Valuation Handbook for Optimized Depreciated Replacement Cost Valuation of System Fixed Assets of Privately Owned Distribution Utilities Operating Under Performance-Based Regulation (Third Regulatory Period)", Energy Regulatory Commission, July 2010.

substation termination equipment, land and easements owned by the DU should be optimized out and the other two 69kV lines should be considered for optimization down to a lower conductor size. In extreme cases it may be necessary to consider supplying consumer demand at a lower operating voltage whilst still meeting with the DU's quality of supply criteria. The notional/optimized network should be based on an achievable and acceptable network design.

B.4 Distribution Substations

Issue (a): Whether the number of distribution substations exceeds that required to meet the DU's quality of supply criteria and projected future load growth within the allowed planning period.

Approach: The location of all distribution substations may be considered fixed. Each substation shall be tested to determine whether a more cost-efficient network would result if distribution substations (and their associated supply/tie-lines) were eliminated whilst still meeting the DU's quality of supply criteria. Network Assets that are not required should be optimized out. For example, a DU's network includes a 69/23kV substation that was built to supply a factory that has closed down. With the loss of the factory, the substation has been reconfigured to supply the surrounding 23kV distribution network but is still lightly loaded. If the 69/23kV substation is removed from service and the 23kV network reconfigured the adjacent 69/23kV substations can comfortably supply the network demand whilst still meeting with the DU's allowed quality of supply criteria. The 69/23kV substation should be optimized out. Again the notional/optimized network should be based on an achievable and acceptable network design. In this situation, it may also be necessary to optimize out the sub-transmission circuits supplying the substation following B.3 above.

Issue (a): Whether the busbar/switchboard arrangements and configurations are required to meet the DU's quality of supply criteria and projected future load growth within the allowed planning period.

Approach: Consider whether double bus and/or double breaker arrangements should be optimized to breaker-and-a-half or single-bus arrangements. In those locations where the redundancy associated with dual busbar/switchboard arrangement is not required optimize out the duplication.

B.5 Primary (MV) Distribution Circuits

Issue (a): Whether the number of primary distribution lines/cables exceeds the number required given the DU's quality of supply criteria and the forecast future load growth within the allowed planning period.

Approach: The route of each line/cable may be considered fixed. Assess the number of primary distribution lines/cables in relation to the allowed quality of supply criteria and future load growth within the allowed planning period. Optimize out those lines/cables that are not required. For example, a DU's network contains a 13.8kV feeder that consists of a lightly loaded line that traverses from one distribution substation to another distribution substation. The feeder/line follows a path that contains other 13.8kV lines and it would be possible to transfer the distribution/line transformers (loads) from the lightly loaded feeder to other lines without exceeding the DU's security criteria (given the load forecast for the allowed planning period). The lightly loaded feeder should be optimized out based on an achievable and acceptable network design.

Another example would be a primary distribution feeder built to supply a large industrial load that has been shut down and over subsequent years a small number of distribution transformers have been connected to the feeder. If these distribution transformers could be more cost-effectively

supplied by short spurs connected to a neighbouring feeder, then the feeder in question (and its associated circuit breaker) should be optimized out and the affected line transformers assumed to be connected to the neighbouring feeder. The value of the short notional spur lines to the Distribution Transformers should be included in the optimized RAB.

Issue (b): Whether three-phase primary distribution lines are required.

Approach: Where an existing distribution line or a part of it is less than three-phase construction, the line shall be valued accordingly. Three-phase primary distribution lateral lines that are in remote/rural areas where consumers have no requirement for three-phase supply shall be optimized to single-phase two-wire lines provided the optimization meets with the DU's quality of supply criteria with a two-wire arrangement. If significant numbers of three-phase laterals exist which could be optimized to single-phase the optimization can be undertaken by sample (i.e. examining a typical network section and determining the percentage of lines for optimization from three-phase to single-phase). This optimization percentage would then be applied across all the relevant part of the network.

Optimization of the Capacity/Voltage of Elements in the DU's Network

B.6 Delivery Point Substations

Issue (a): Whether all the equipment located at the delivery point substations is required, given the allowed quality of supply criteria.

Approach: Optimize the size of the equipment used, including transformers, to the lowest standard rating that meets the accepted quality of supply criteria coupled with the projected future load growth for the relevant planning period.

B.7 Sub-transmission (HV) Lines and Cables

Issue (a): Conductor and cable size and voltage.

Approach: Determine the required capacity and/or voltage of the line/cable given the allowed quality of supply criteria and the predicted load growth within the allowed planning period. Optimize down the size/voltage of the conductor or cable to the most cost-efficient standard size/voltage whilst still meeting the allowed quality of supply criteria. If possible, use the short-term line/cable ratings as appropriate. For example, a remote urban area is supplied via a 138/23kV substation which in turn is supplied via a single circuit 138kV line from a Delivery Point Substation that has both 138kV and 69kV assets. The 138kV supply line well exceeds that required to supply the projected urban area load within the planning period. It would be possible to supply the projected load whilst still meeting the allowed quality of supply criteria using a more economic 69kV single circuit line and associated 69/23kV substation. For valuation purposes, the existing supply network should be optimized down to a notional 69kV line coupled with a 69/23kV substation.

Issue (b): Whether underground cables are justified.

Approach: Review the existing underground sub-transmission to determine whether underground reticulation is justified.

Possible justifications for undergrounding include:

i. local authority planning criteria prohibit the construction of new overhead circuits;

ii. the use of underground cable is the most cost-efficient means of achieving the disclosed quality of supply criteria;
iii. economic analysis shows that underground cable is the most cost-efficient method of providing the required network service;
iv. consultation with customers affected (including those affected by having to pay higher electricity distribution prices) has demonstrated a willingness to pay the additional cost of the underground service; or
v. the existing underground cable was funded by a capital contribution equal to, as a minimum, the difference in the capital cost of overhead and underground circuits.

If suitable justification for the existing underground cables cannot be provided, then optimize the underground circuits to overhead. The justification for retaining underground cable in the optimized network shall be described in general terms in the Independent Expert's Report.

B.8 Distribution Substations

Issue (a): Under-utilized equipment installed at substations.

Approach: Optimize the size of the equipment used, including transformers, to the lowest standard rating that meets the accepted quality of supply criteria coupled with the projected future load growth for the accepted planning period. For example, a Distribution Substation is equipped with a single 40MVA transformer, whilst the peak projected demand over the planning period is 18MVA. The use of a 20MVA unit would meet the DU's quality of supply criteria, and the DU has 20MVA units installed in other substations. The 40MVA transformer should be optimized down to 20MVA.

Issue (b): Over-rated equipment installed at substations.

Approach: Optimize the nominal and fault ratings of the equipment used in substations to the lowest standard rating that meets the accepted quality of supply criteria coupled with the projected future load growth and network developments over the accepted planning period. For example, a substation is equipped with 40kA fault rated equipment and the fault level is projected to increase only marginally within the accepted planning period. The DU widely uses 18kA fault rated equipment and it would be adequate for the planning period. The relevant substation should be optimized down to 18kA fault rated MEA.

B.9 Primary Distribution (MV) Lines and Cables

Issue (a): Conductor and cable size and voltage.

Approach: Determine the required capacity and/or voltage of the primary distribution lines/cables given the allowed quality of supply criteria and the predicted load growth within the allowed planning period. Optimize down the size/voltage of the conductor or cable to the most cost-efficient standard size/voltage whilst still meeting the allowed quality of supply criteria. If possible, use the short-term line/cable ratings as appropriate. For example, the primary distribution feeders associated with a distribution substation are lightly loaded. The distribution lines can be optimized to a smaller conductor size (i.e. from 336MCM to 4/0) without breaching the DU's primary distribution line quality of supply criteria (given the forecast demand for the allowed primary distribution line planning period). The primary distribution lines should be optimized down unless the higher capacity can be justified because that it is required to backup an adjacent Distribution Substation or primary distribution feeder in the event of a contingency situation arising.

Issue (b): Whether underground cables are justified.

Approach: Review the existing underground primary distribution cables to determine whether underground reticulation is justified.

Possible justifications for undergrounding include:

i. local authority planning criteria prohibit the construction of new overhead circuits;
ii. the use of underground cable is the most cost-efficient means of achieving the disclosed quality of supply criteria;
iii. economic analysis shows that underground cable is the most cost-efficient method of providing the required network service;
iv. consultation with customers affected (including those affected by having to pay higher electricity distribution prices) has demonstrated a willingness to pay the additional cost of the underground service; or
v. health and safety reasons.

If there is clear evidence that underground cables cannot be justified, then optimize them to overhead lines.

Issue (c): Underground cable trenching.

Approach: Optimize the trenching arrangement of existing underground cables. Cables running close together, or on the same side of any road or street shall be optimized to a single trench except where this would not meet the quality of supply criteria. Derating factors applicable to cables run in a single trench should be considered when making this assessment. If more than one underground cable is laid in a trench, only the incremental cost of the additional cable(s) may be included in the valuation.

Issue (d): Overhead lines constructed along the same street.

Approach: Optimize the overhead arrangement of existing overhead lines. Overhead lines running close together, or on the same side of any road or street shall be optimized to a double circuit or under-built line except where this would meet the quality of supply criteria. For under-built lines only the incremental cost of the additional conductor and pole-top hardware may be included in the valuation. For example, due to different construction timings, a new 23kV overhead line has been installed adjacent to an existing 23kV overhead distribution line. The most economical configuration is a single-pole line supporting both 23kV circuits. The two existing lines should be optimized to a notional double circuit 23kV line.

B.10 Distribution Transformers (pole, kiosk, ground types)

Issue: Utilization of transformer capacity.

Approach: Optimize out excess distribution transformer capacity so that the capacity utilization is at a level judged to be efficient for the DU, given its load and supply characteristics. If maximum demand information is available for individual distribution transformers they should be individually optimized to the lowest possible standard size, whilst considering the transformer's projected demand for the allowed planning period. This approach should be applied to all DU owned transformers supplying individual customers.

For those distribution transformers where individual maximum demands are not available, the optimization should be based on transformer capacity utilization which is defined as follows:

$$\text{Utilization (\%)} = \frac{\text{Peak Demand (MVA)} \times 100}{\text{Total Distribution Transformer Capacity (MVA)}}$$

If a DU's transformer utilization level is less than 30%, then distribution transformer capacity should be optimized out such that the optimized network has a distribution utilization level that is greater than 30%. Any distribution transformer capacity adjustments that are made shall be disclosed and the methodology used clearly explained in the Independent Expert's Report. Transformer capacity that is optimized out shall be valued at the average depreciated replacement cost (DRC) per kVA of the DU's transformer equipment.

DUs may separate out segments of the network and apply the above formula (i.e. distribution (MV) feeders). Again, if this approach is taken, details shall be included in the optimization description included in the Independent Expert's Report.

When using the above formula, care should be taken to:

i. separate out the capacity associated with privately-owned distribution transformers;
ii. ensure that the peak demand value used includes only the electrical load which passes through the distribution transformers being analyzed. The need to remove load that does not pass through the distribution transformers may mean that the peak load used in the analysis has to be estimated, and details of the analysis used to arrive at the peak load estimate should be included in the Independent Expert's Report; and
iii. account for embedded generation that would reduce peak network demand levels.

B.11 Secondary (LV) Distribution

Issue (a): Whether secondary distribution underground cables are justified.

Approach: Review existing underground secondary distribution (LV) cables to determine whether undergrounding is justified. Possible justifications for undergrounding include:

i. local authority planning criteria prohibit the construction of new overhead circuits;
ii. the use of underground cable is the most cost-efficient means of achieving the accepted quality of supply criteria;
iii. economic analysis shows that underground cable is the most cost-efficient method of providing the required network service;
iv. consultation with customers affected (including those affected by having to pay higher electricity distribution prices) has demonstrated a willingness to pay the additional cost of the underground service; or
v. health and safety reasons.

If there is clear evidence that underground cables cannot be justified, then optimize them to overhead lines.

Issue (b): Underground primary distribution trenching.

Approach: Optimize the trenching arrangement of existing underground cables. Cables running close together, or on the same side of any road or street shall be optimized to a single trench except where this would not meet the accepted quality of supply criteria. Derating factors applicable to cables run in a single trench should be considered when making this

assessment. If more than one underground cable is laid in a trench only the additional cables' incremental cost may be included in the valuation.

Optimization of Network Engineering

B.12 System Control and Communication

Issue (a): Degree of sophistication of SCADA equipment.

Approach: Determine whether the equipment is appropriate based on disclosed Quality of Supply criteria. Reduce replacement cost to that of a MEA of the required sophistication. Consideration should also be made regarding the optimization of control centres. For example, a DU consists of the amalgamation of six historical networks that include the associated original six control centres. The MEA would be a notional single control centre with an associated notional emergency backup control centre. There may be reasons to maintain multiple active control centres (i.e. for security reasons). If this is the case, the reasons should be clearly outlined in the Independent Expert's Report. The use of fibre/communication networks for system control should be limited to the amount required to control/operate the electrical network. Underutilized fibre should be optimized out.

B.13 Distribution Substations

Issue (a): Land and buildings.

Approach: Optimize indoor substations to outdoor where land is available and where this will result in a more cost-efficient design unless there are clear technical or security reasons or local authority requirements that prevent this. Optimize out any unutilized or under-utilized land so that the value of the land allowed reflects only the area of land required to meet the accepted quality of supply criteria and projected future load growth for the accepted planning period. Reduce the replacement cost of buildings to that of a simple standard modern structure cost-effective design. A higher standard of construction is allowed only where the DU can provide evidence to show that a lower-cost design will not meet local authority planning requirements, given substation's location. The size of the optimized design should not exceed that required to meet the building's essential functions.

Issue (b): Whether substation engineering exceeds DU requirements.

Approach: Review the standard of engineering of each substation. If possible, recent projects undertaken by the DU should be used as a benchmark for this test. If it is found that a more cost-efficient standard of engineering would meet the accepted quality of supply criteria, the existing assets should be notionally re-engineered and the replacement costs reduced accordingly. Compliance with territorial local authority conditions for the substation location should be retained in any notional redesign.

Issue (c): Fire protection and oil retention facilities.

Approach: Include equipment currently installed unless not required for MEA. Again recent projects undertaken by the DU should be used as a benchmark for this test.

B.14 Secondary (LV) Distribution Lines/Cables

Issue (a): Whether the secondary distribution network's engineering exceeds the standard required to meet the accepted quality of supply criteria.

Approach: Review the standard of engineering of the secondary/LV distribution network, using recent projects undertaken by the DU as a benchmark for this test. If it is found that a more cost-effective standard of engineering would meet the accepted quality of supply criteria, those parts of the low voltage network containing excess asset value should be notionally reconfigured so that they do not exceed the required standard. Assets that are not required should be optimized out. In applying this test, it is not required that DUs examine each individual primary/LV circuit. It is acceptable to estimate the proportion of the DU's low voltage distribution network that is over-engineered and apply an appropriate optimization factor. However, details of the approach taken shall be included in the optimization description included in the Independent Expert's Report.

B.15 Voltage Control Devices

Issues: (i) Degree of voltage control, (ii) Manual and on-load tap changes, (iii) Line regulators and line drop compensation (iv) Reactive compensation.

Approach: Test requirements for all existing voltage control devices and optimize out where there is no clear justification for the equipment. For example, a DU's distribution substation is equipped with a single 40MVA transformer bank with an Off Load Tap Changer (OLTC). The 13.8kV feeders are all equipped with line regulators adjacent to the substation. A more cost-effective design involves the use of a 40MVA transformer bank equipped with OLTC and the DU has recently installed a Distribution Substation Transformer with OLTC. If cost-effective the line regulators should be optimized out and replaced with a notional 40MVA bank with OLTC capability.

B.16 Network Spares

Issue: The extent/quantity of network-related spares.

Approach: An assessment should be made of network spares. Those spares which are of the wrong type, or which for whatever reason are unlikely to be used by the DU within the relevant planning horizon, should be optimized out. Spares would likely be classified as (i) critical contingency spares (for N-1 security), (ii) emergency spares and (ii) routine spares (for maintenance). The extent of the spares will be dictated by (i) equipment procurement lead times and (ii) asset failure rates and (iii) network reliability requirements. Network spares should not be dictated by the DU's inventory associated with capital works. The stores history about spares shall be taken into account when assessing reasonable levels of spares.

APPENDIX C: UPDATED REPLACEMENT COSTS

C.1 Introduction

This Appendix provides the updated standard asset replacement costs.

C.2 Purpose of the Replacement Cost Update

Clause 4.8 of the RDWR[42] describes the requirements for the valuation of the RAB of DUs:

a) The RAB for a Regulated Distribution System is re-valued; or
b) The previous value of the RAB for a Regulated Distribution System is rolled forward.

Regardless of which of the above options applies to an entry group, the asset valuation shall be undertaken in accordance with the requirements of this Handbook.

The updated replacement costs presented in this Appendix C is to be utilized by the Commission as they deem appropriate for purposes of supporting the next Regulatory Period.

C.3 Replacement Cost – SYSTEM FIXED ASSETS

For this replacement cost update, the internationally accepted principle of absolute valuation by replacement cost analysis, where required absolute valuation by modern equivalent asset (MEA) analysis, was used to develop the standard replacement costs of System Fixed Assets as shown in this Appendix C.

The replacement cost of all individual System Fixed Assets is derived from the lowest cost to replace an existing asset with a MEA with an equivalent service potential and capacity. Furthermore, the MEA should be commercially available and should not require re-engineering or redesign of the existing asset.

For commonly used System Fixed Assets, the standard replacement costs for MEAs are listed in this Appendix C. These costs are based on December 31, 2019 costs and escalation of standard replacement costs would have to be applied for years beyond 2019 by using the Consumer Price Index (CPI). A valuation report must state the indexation factor used to inflate the standard replacement costs provided in this Appendix C and include an analysis of how this indexation factor was derived. The standard replacement costs should be revaluated prior to each reset of the First Entry Group.

The MEAs as listed in this Appendix C includes costs for all materials[43], labour[44], plant[45] and overheads[46].

---

[42] ERC Case No. 2019-003 RM Annex "A", "Rules for Setting Distribution Wheeling Rates for Privately Owned Electricity Distribution Utilities Operating Under Performance Based Regulation", Energy Regulatory Commission, December 2021.

[43] Costs of materials delivered to store inclusive of any taxes paid excluding VAT.

[44] Costs of direct labour including survey, design and construction and labour on-costs incorporating an allowance for holiday pay, sick leave, training, other unproductive time, workers compensation payments, superannuation, and payroll tax.

[45] Costs associated with transport and plant for delivery and erection.

[46] Costs associated with preparation of master-plans, use of consultants, capital rationing processes, project approval and budgeting, contract administration, site supervision and construction related corporate administrative overhead, asset register data processing including new asset data input, retirements for replaced assets and associated procedures, to the extent that these costs relate directly to the RAB.

For System Fixed Assets not listed in this Appendix C, the valuer is to determine the MEA for these non-standard assets, including the efficient replacement cost and the standard asset life. This will be based on an asset that:

a) has an equivalent service potential to the existing asset;
b) can be constructed or purchased at the time of the valuation with current technology; and
c) has the lowest lifetime cost.

In developing the replacement cost for non-standard assets, a 'Brownfields' cost basis should be adopted. Also, the replacement cost should correspond to a significant scale of construction and not a gradual addition.

For non-standard asset replacement costs, equipment purchase costs[47] should be based on quotations from suppliers or manufacturers operating in a competitive market and labour[48], overhead[49] and plant[50] costs will be based on competitive costs based on efficient industry practice.

[47] Supra note 84.

[48] Supra Note 85.

[49] Supra note 86.

[50] Supra note 87.

STANDARD REPLACEMENT COSTS FOR DISTRIBUTION TRANSFORMERS

Asset Class	Unit	Notes	Standard Value (PhP '000)	Standard Value (PhP '000)	Standard Value (PhP '000)	Standard Life (Years)
			Single Transformer	Two Transformers	Three Transformers	
Pole Mounted, Single Phase, 4.8kV Primary Voltage, 480V and lower Secondary Voltage						
25kVA	No.	b	101	178	*	30
50kVA	No.	b	142	*	*	30
Pole Mounted, Single Phase, 6.24/3.6kV Primary Voltage, 480V and lower Secondary Voltage						
25kVA	No.	b	100	*	*	30
50kVA	No.	b	142	267	403	30
75kVA	No.	b	176	331	503	30
100kVA	No.	b	217	409	623	30
167kVA	No.	b	*	*	948	30
Pole Mounted, Single Phase, up to 7.62kV Primary Voltage, 480V and lower Secondary Voltage						
up to 10kVA	No.	b	79	134	215	30
15kVA	No.	b	88	148	275	30
25kVA	No.	b	100	189	282	30
37.5kVA	No.	b	116	228	364	30
50kVA	No.	b	137	261	377	30
75kVA	No.	b	177	324	554	30
100kVA	No.	b	229	432	747	30
167kVA	No.	b	282	518	827	30
250kVA	No.	b	326	*	1255	30
333kVA	No.	b	408	*	1427	30
Pole Mounted, Single Phase, 13.2 or 13.8kV Primary Voltage, 480V and lower Secondary Voltage						
up to 10kVA	No.	b	81	163	211	30
15kVA	No.	b	98	178	236	30
25kVA	No.	b	105	184	266	30
37.5kVA	No.	b	119	210	317	30
50kVA	No.	b	138	248	369	30
75kVA	No.	b	174	321	432	30
100kVA	No.	b	229	432	657	30
167kVA	No.	b	282	518	827	30
250kVA	No.	b	334	*	*	30
333kVA	No.	b	418	*	*	30
Pole Mounted, Single Phase, 19kV and above Primary Voltage, 480V and lower Secondary Voltage						
up to 15kVA	No.	b	92	174	268	30
25kVA	No.	b	88	201	310	30
37.5kVA	No.	b	98	223	343	30
50kVA	No.	b	119	250	384	30
75kVA	No.	b	143	352	505	30
100kVA	No.	b	181	381	605	30
167kVA	No.	b	242	599	922	30
250kVA	No.	b	301	*	1246	30
333kVA	No.	b	336	*	*	30
Pole Mounted, Single Phase, 34.5kV and above Primary Voltage, 480V and lower Secondary Voltage						
up to 15kVA	No.	b	92	174	268	30
25kVA	No.	b	97	204	260	30
37.5kVA	No.	b	101	212	276	30
50kVA	No.	b	136	261	363	30
75kVA	No.	b	150	289	424	30
100kVA	No.	b	218	370	536	30
167kVA	No.	b	292	586	775	30
250kVA	No.	b	383	*	1218	30
333kVA	No.	b	452	*	1508	30
Pad mounted 1Phase Transformer, 13.2 or 13.8kV Primary Voltage						

Asset Class	Unit	Notes	Standard Value (PhP '000)	Standard Value (PhP '000)	Standard Value (PhP '000)	Standard Life (Years)
			Single Transformer	Two Transformers	Three Transformers	
75kVA	No.	a	233	*	*	30
167kVA	No.	a	*	511	749	30
250kVA	No.	a	*	771	1139	30
333kVA	No.	a	*	923	1367	30
Pad mounted 3Phase Transformer, 13.2 or 13.8kV Primary Voltage						
300kVA	No.	a	976	*	*	30
500kVA	No.	a	1195	*	*	30
750kVA	No.	a	1332	*	*	30
1000kVA	No.	a	1453	*	*	30
1500kVA	No.	a	1616	*	*	30
2000kVA	No.	a	2659	*	*	30
Pad mounted 1Phase Transformer, 34.5kV Primary Voltage						
75kVA	No.	a	314	*	*	30
167kVA	No.	a	446	*	*	30
Pad mounted 3Phase Transformer, 34.5kV Primary Voltage						
75kVA	No.	a	950	*	*	30
150kVA	No.	a	950	*	*	30
300kVA	No.	a	1096	*	*	30
500kVA	No.	a	1394	*	*	30
750kVA	No.	a	1512	*	*	30
1000kVA	No.	a	1674	*	*	30
1500kVA	No.	a	1862	*	*	30
2000kVA	No.	a	2050	*	*	30

Note:
a - Single three phase transformer. Includes all costs to establish transformer (i.e., equipment, installation, plant and overhead costs) including HV fuses and surge arrestors.
b - Includes all costs to establish transformer (i.e., equipment, installation, plant and overhead costs) including HV fuses and surge arrestors.
* - Non-Standard
Asset
** - Not Applicable

STANDARD REPLACEMENT COSTS FOR OVERHEAD LINES: CONDUCTORS

Asset Class	Unit	Notes	Standard Value (PhP '000)	Standard Life (Years)
Aluminium Bare or ACSR - 3 wires				
795 MCM Bare ACSR Conductor	km	a	456	30
556 MCM Bare ACSR Conductor	km	a	354	30
477 MCM Bare ACSR Conductor	km	a	261	30
397.5 MCM Bare ACSR Conductor	km	a	214	30
336.4 MCM Bare ACSR Conductor (30/7)	km	a	229	30
336.4 MCM Bare ACSR Conductor (26/7)	km	a	189	30
#4/0 AWG Bare ACSR Conductor	km	a	128	30
#3/0 AWG Bare ACSR Conductor	km	a	103	30
#2/0 AWG Bare ACSR Conductor	km	a	88	30
#1/0 AWG Bare ACSR Conductor	km	a	79	30
#2 AWG Bare ACSR Conductor	km	a	51	30
#4 AWG Bare ACSR Conductor	km	a	44	30
#6 AWG Bare ACSR Conductor	km	a	42	30
Aluminium Insulated or ACSR (Aerial)				
477 MCM AWG Insulated ACSR Cable	km	a	269	30
397.5 MCM AWG Insulated ACSR Cable	km	a	226	30
336.4 MCM AWG Insulated ACSR Cable	km	a	265	30
#4/0 AWG Insulated ACSR Cable	km	a	151	30
#3/0 AWG, Insulated ACSR Cable	km	a	102	30
#2/0 AWG, Insulated ACSR Cable	km	a	87.5	30
#1/0 AWG Insulated ACSR Cable	km	a	78.0	30
#2 AWG Insulated ACSR Cable	km	a	50.4	30
#4 AWG Insulated ACSR Cable	km	a	43.8	30
#6 AWG Insulated ACSR Cable	km	a	41.3	30
Copper Bare				
#4/0 AWG Bare Copper Conductor	km	a	500	30
#2/0 AWG Bare Copper Conductor	km	a	302	30
#1/0 AWG Bare Copper Conductor	km	a	259	30

Note:
a - Applies to single conductor and includes all costs to erect/establish/string a single overhead conductor, but excludes the costs associated with poles and pole top hardware.

STANDARD REPLACEMENT COSTS FOR OVERHEAD LINES: POLES

Asset Class	**Unit**	**Notes**	**Standard Value** (PhP '000)	**Standard Life** (Years)
Wooden Poles				
7.5 M or 8.0 M (25 FT)	No.	a	10.1	20
9.0 M or 9.5 M (30 FT)	No.	a	12.8	20
10.5 M or 11M (35 FT)	No.	a	17.0	20
12.0 M (40 FT)	No.	a	25.7	20
13.5 M (45 FT)	No.	a	36.3	20
15.0 M (50 FT)	No.	a	50.3	20
16.5 M (55 FT)	No.	a	65.0	20
18.0 M (60 FT)	No.	a	87.0	20
19.5 M (65 FT)	No.	a	108.9	20
21.0 M (70 FT)	No.	a	122.5	20
22.5 M or 23.0 M (75 FT)	No.	a	136.0	20
Concrete Poles				
7.5 M or 8.0 M (25 FT)	No.	a	13.8	30
9.0 M (30 FT)	No.	a	16.8	30
9.5 M (32 FT)	No.	a	20.5	30
10.5 M or 11M (35 FT)	No.	a	24.2	30
12.0 M (40 FT)	No.	a	27.3	30
13.5 M (45 FT)	No.	a	33.2	30
15.0 M (50 FT)	No.	a	40.3	30
16.5 M (55 FT)	No.	a	42.5	30
18.0 M (60 FT)	No.	a	56.3	30
20.0 M (65 FT)	No.	a	99.5	30
21.5 M (70 FT)	No.	a	145.8	30
23.0 M (75 FT)	No.	a	167.9	30
24.5 M (80 FT)	No.	a	214.4	30
27.5M, (90 FT)	No.	a	248.9	30
Steel Poles				
7.5 M or 8.0 M (25 FT)	No.	a	15.0	40
9.0 M or 9.5 M (30 FT)	No.	a	20.3	40
10.5 M or 11M (35 FT)	No.	a	27.4	40
12.0 M (40 FT)	No.	a	37.6	40
13.5 M (45 FT)	No.	a	45.3	40
13.5 M or 15.0 M (50 FT)	No.	a	58.0	40
16.5 M (55 FT)	No.	a	105.2	40
18.0 M (60 FT)	No.	a	148.6	40
20.0 M (65 FT)	No.	a	169.4	40
21.0 M (70 FT)	No.	a	202.2	40
22.5 M or 23M (75 FT)	No.	a	239.1	40
24.0 M (80 FT)	No.	a	301.3	40
24.0 M (85 FT)	No.	a	328.8	40
27.5 M (90 FT)	No.	a	353.3	40
27.5 M or 29 M (95 FT)	No.	a	538.7	40
32 M (105 FT)	No.	a	1192.3	40
Galvanised Iron				
20 FT	No.	a	5.4	30

Note:
a - Includes all costs to erect/establish poles and the associated support structures. The costs include earth rods and mats but exclude aerial earth wires.

STANDARD REPLACEMENT COSTS FOR OVERHEAD LINES: POLE TOP HARDWARE

Asset Class	Unit	Notes	Standard Value (PhP '000)	Standard Value (PhP '000)	Standard Value (PhP '000)	Standard Life (Years)
			Straight, Light to Medium Angle(0-60 deg)	Corner (60-90 deg)	Dead End	
One Wire						
7.65kV	No.	a	1.3	3.1	3.6	30
13.2 or 13.8kV	No.	a	1.4	3.4	4.1	30
Single Phase (Two Wire)						
240-480V	No.	a	1.4		0.7	30
7.65kV	No.	a	4.7	22.4	11.6	30
13.2 or 13.8kV	No.	a	7.7	25.3	13.0	30
Three Phase						
240-480V	No.	a	2.9		1.6	30
7.65kV	No.	a	6.8	26.6	17.3	30
13.2kV or 13.8kV	No.	a	9.6	30.0	19.5	30
19kV	No.	a	11.0	34.5	22.4	30
34.5kV	No.	a	13.9	40.4	24.7	30
69kV	No.	a	39.8	62.0	49.2	30
115kV	No.	a	134.4	166.2	105.2	30
138kV	No.	a	134.4	166.2	105.2	30

Note:
a - Includes all the pole top hardware required to support the overhead conductor, but excludes the pole structure. Double dead end construction can be accounted for by using two dead end Standard Values, the costs includes any suspension and post insulators.

STANDARD REPLACEMENT COSTS FOR CUSTOMER SERVICE CONNECTIONS

Asset Class	Unit	Notes	Standard Value (PhP '000)	Standard Life (Years)
Copper Cable, 600V				
Single 1000MCM	m	a	2.931	30
Single 750MCM	m	a	2.522	30
Single 500MCM	m	a	1.619	30
Single 400MCM	m	a	1.301	30
Single 350MCM	m	a	0.984	30
Single 300MCM	m	a	0.686	30
Single 250MCM	m	a	0.650	30
Single 125MCM	m	a	0.646	30
Single #4/0	m	a	0.643	30
Single #3/0	m	a	0.553	30
Single #2/0	m	a	0.448	30
Single #1/0	m	a	0.405	30
Single #2	m	a	0.265	30
Triplex #2	m	a	0.616	30
Quadruplex #2	m	a	0.751	30
Single #4	m	a	0.206	30
Single #6	m	a	0.157	30
Single #8	m	a	0.066	30
Single #10	m	a	0.052	30
Single #12	m	a	0.035	30
Single #14	m	a	0.030	30
Duplex #8	m	a	0.127	30
PDX #10	m	a	0.198	30
PDX #12	m	a	0.168	30
Aluminium Cable, 600V				
Single #1/0	m	a	0.107	30
Single #2	m	a	0.076	30
Single #4	m	a	0.059	30
Single #6	m	a	0.029	30
Single #8	m	a	0.027	30
Duplex #6	m	a	0.050	30
Duplex #8	m	a	0.057	30

Notes:

a - The costs are per meter to install a customer service connection including all accessories, but excludes the costs associated with meters and relays.

STANDARD REPLACEMENT COSTS FOR METERS AND METERING ACCESSORIES

Asset Class	Unit	Notes	Standard Value (PhP '000)	Standard Life (Years)
Meters				
Single Phase Class 100 (ElectroMechanical)	No.	b	2.5	25
Single Phase Class 100 (Electronic)	No.	b	3.3	15
Single Phase Class 200 (ElectroMechanical)	No.	b	3	25
Single Phase Class 200 (Electronic)	No.	b	2	15
Three Phase Class 200 (Electromechanical Self-Contained)	No.	b	10.2	15
Three Phase Class 200 (Electronic Self-Contained)	No.	b	12.6	15
Single Phase Electromechanical CT meter including CT	No.	b, c	21.9	25
Single Phase Electronic CT meter including CT	No.	b, c	21.3	25
Three Phase Electronic CT meter including CT	No.	a, c	30	25
Current Transformers				
69 kV single or two core	No.		331	25
34.5 kV single core	No.		94.3	25
13.2kV or 13.8 kV single core	No.		75.5	25
600 V single core (200A)	No.		11.3	25
600 V single core (800A)	No.		16.1	25
Potential Transformers				
115kV	No.		348.7	25
69 kV	No.		222.2	25
34.5kV	No.		69.2	25
13.8 kV	No.		57.9	25

Notes
a - Three Phase Unit cost. Includes installation, plant and overhead costs for meter and accessories required for installation.
b - Single Phase Unit cost. Includes installation, plant and overhead costs for meter and accessories required for installation.
c - The standard life for electromechanical meters are listed in the table, however a standard life of 15 years could be applied for electronic meter of this type.

STANDARD REPLACEMENT COSTS FOR OVERHEAD LINE DEVICES

Asset Class	Unit	Notes	Standard Value (PhP '000)	Standard Life (Years)
Capacitors				
Single Phase Unit, Line, 13.2kV or 13.8kV, 200kVAr	No.	a	61.5	30
Single Phase Unit, Line, 13.2kV or 13.8kV, 100kVAr	No.	a	47.0	30
Single Phase Unit, Line, 13.2kV or 13.8kV, 50kVAr	No.	a	35.4	30
Single Phase Unit, Line, 34.5kV, 200kVAr	No.	a	83.2	30
Lightning Arresters				
Single Phase Surge Arrester, Station Class, 230kV	No.	a	118.8	30
Single Phase Surge Arrester, Station Class, 115kV	No.	a	61.8	30
Single Phase Surge Arrester, Station Class, 69kV	No.	a	67.3	30
Single Phase Surge Arrester, Station Class, 34.5kV	No.	a	34.8	30
Single Phase Surge Arrester, Station Class, 23kV	No.	a	33.8	30
Single Phase Surge Arrester, Station Class, <=13.8kV	No.	a	33.0	30
Single Phase Surge Arrester, Distribution Class, 34.5kV	No.	a	4.7	30
Single Phase Surge Arrester, Distribution Class, 23kV or 24kV	No.	a	4.7	30
Single Phase Surge Arrester, Distribution Class, <=13.8kV	No.	a	4.2	30
Fuses				
Cut-out Fuse, Distribution Type, 7.8 kV & below	No.	a	3.7	30
Cut-out Fuse, Distribution Type, Rated 15 kV 100A	No.	a	4.8	30
Cut-out Fuse, Distribution Type, Rated 15 kV 200A	No.	a	8.4	30
Cut-out Fuse, Distribution Type, Rated 23 kV 100A	No.	a	6.0	30
Cut-out Fuse, Distribution Type, Rated 27 kV 100A	No.	a	6.0	30
Cut-out Fuse, Distribution Type, Rated 34.5 kV	No.	a	9.6	30
Load Break Fuse Cut-out, Rated 15 kV 100A, 3 phase	No.	a	16.2	30
Load Break Fuse Cut-out, Rated 15 kV 200A, 3 phase	No.	a	19.5	30
Switches, Circuit Breakers and Disconnectors				
Single Phase Disconnect Switch, Rated 15 kV 600 A	No.	a	17.7	30
Fault Indicators				
Fault Indicator, 13.8 kV	No.	a	23.4	30
Fault Indicator, 23 kV	No.	a	23.5	30
Fault Indicator, 34.5 kV	No.	a	29.4	30
Reclosers				
Three Phase Circuit Recloser, Rated 34.5KV 560A or 630A	No.	a	875.3	30
Three Phase Circuit Recloser, Rated 15KV 630A	No.	a	820.3	30
Three Phase Circuit Recloser, Rated 15KV 560A	No.	a	605.7	30

Notes

a - Includes installation, plant and overhead costs for unit and accessories required for installation.

STANDARD REPLACEMENT COSTS FOR SWITCHES AND CIRCUIT BREAKERS

Asset Class	Rated Voltage (kV)	Rated (Ampere)	Location	Unit	Notes	Standard Value (PhP '000)	Standard Life (Years)
Switches, Circuit Breakers and Disconnectors							
Three Phase Disconnect Switch, Station Class	115	2000		No.	a	689.2	30
Three Phase Disconnect Switch, Station Class	69	1200		No.	a	623.3	30
Three Phase Disconnect Switch, Subtransmission Type	69	1200		No.	a	659.6	30
Three Phase Motorized Disconnect Switch, Station Class	15	2000		No.	a	573.6	30
Three Phase Motorized Disconnect Switch, Station Class	15	1200		No.	a	500.4	30
Three Phase Motorized Disconnect Switch, Station Class	15	2000		No.	a	259.9	30
Single Phase Motorized Disconnect Switch, Station Class	15	1200		No.	a	221.0	30
Circuit Breakers							
Gas Power Circuit Breaker, Dead Tank	13.8	2000	Outdoor	No.	a	1,309.2	30
Gas Power Circuit Breaker, Dead Tank	13.8	3150	Outdoor	No.	a	3,218.3	30
Gas Power Circuit Breaker, Dead Tank	34.5	1200	Outdoor	No.	a	2,584.5	30
Gas Power Circuit Breaker, Dead Tank	34.5	2000	Outdoor	No.	a	2,630.4	30
Gas Power Circuit Breaker, Dead Tank	69	1200	Outdoor	No.	a	2,615.6	30
Gas Power Circuit Breaker, Dead Tank	69	2500	Outdoor	No.	a	3,059.7	30
Gas Power Circuit Breaker, Dead Tank	69	3150	Outdoor	No.	a	1,390.8	30
Gas Power Circuit Breaker, Dead Tank	115	2000	Outdoor	No.	a	3,862.1	30
Gas Power Circuit Breaker, Dead Tank	115	1200	Outdoor	No.	a	2,844.7	30
Gas Power Circuit Breaker, Dead Tank	230	2000	Outdoor	No.	a	6,102.9	30
Gas Power Circuit Breaker, Live Tank	115	2000	Outdoor	No.	a	2,324.2	30
Gas Power Circuit Breaker, Live Tank	115	3150	Outdoor	No.	a	2,567.3	30
Air Insulated (Metal Clad)	34.5	1600	Outdoor	No.	a	12,747.2	30
Air Insulated (Metal Clad)	13.8	1600	Outdoor	No.	a	7,403.7	30
Gas Insulated	34.5	1600	Indoor	No.	a	24,456.1	30
Gas Insulated	115	2000	Indoor	No.	a	74,533.6	30
Vacuum Circuit Breaker	15	1200/1250	Outdoor	No.	a	616.0	30

Notes

a - Includes installation, plant and overhead costs for unit and accessories required for installation.

STANDARD REPLACEMENT COSTS FOR POWER TRANSFORMERS

Asset Class	Prim kV	Sec kV	Rating (MVA)	Unit	Notes	Standard Value (PhP '000)	Standard Life (Years)
Power Transformers							
Power Transformer	230	115	300	No.	a	102,635	40
Power Transformer	230	69	150	No.	a	101,483	40
Power Transformer	115/110	34.5	83	No.	a	53,241	40
Power Transformer	115/110	13.8	33	No.	a	39,184	40
Power Transformer	69	13.2 or 13.8	33	No.	a	24,397	40
Power Transformer	69	13.2 or 13.8	21/28	No.	a	22,508	40
Power Transformer	69	13.2 or 13.8	20	No.	a	15,937	40
Power Transformer	69	13.2 or 13.8	15/20	No.	a	14,910	40

Notes

a – Includes transportation, installation, plant and overhead costs for unit and accessories required for installation.